

MAGNETOM

Replacement of Parts System

RF System

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR:
from Impact/Expert
from Vision

Document Version

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1.1 Adjustments following Component Replacement



Ensure that the system is powered on for at least 20 minutes before performing any measurements and/or Service Software procedures. This is to stabilize the temperature in the control cabinet.



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. RFPA) and the protective earth wire connected to the "main component" (e.g. ACC).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

1.1.1 RFSU

Assembly	Adjustment required	Final checks
TX-Module	Perform TuneUp: ■ Tuning Calibration ■ RF Characteristic	Perform Quality Assurance: ■ CoilCheck(SN) ■ Calc Artefacts
RX-Module	Perform TuneUp: ■ Tuning Calibration ■ RF Characteristic ■ Receive Path Calibration	Perform Quality Assurance: ■ CoilCheck(SN) ■ Receive Path Calibration ■ Calc Artefacts
Small signal RF-cable	Perform TuneUp: ■ Tuning Calibration	Perform Quality Assurance: ■ CoilCheck(SN)



When replacing the TX/RX-Modules, carefully and without use of force push in the new TX/RX-Modules
Otherwise you may damage the connectors of the modules

1.1.2 RFPA

**CAUTION**

There is the risk to damage the RFPA by switching it via circuit breaker F8 in the CCA cabinet.

- » Never use the circuit breaker F8 in the CCA cabinet to switch the RFPA (DORA) ON or OFF, if the MR system is powered ON.

**CAUTION**

There is the risk of irreversible destruction of the device by freezing up of the internal water pipes during shipping.

Before returning a replaced RFPA (DORA/or 8kW RFPA), drain the water from the pipes of the device by opening both ends

- » Collect the water/cooling fluid and use it for refilling the cooling system.

**CAUTION**

Never unplug the quick connectors from the RFPA (DORA/or 8kW RFPA) when the water pump in the RCA is in operation (because of high static pressure).

- » Switch off the water pump via the circuit breaker F1 in the RCA cabinet.

NOTICE

Observe the delivered instruction for handling the FOC's when replacing the RFPA:

- » Remove the FOC's at the RFPA carefully, pulling them by hand straight away from the connectors without tilting.
- » Never use a tool like a screwdriver etc. as a lever to stem the double connectors out.

**WARNING**

High voltage!

Hazard of electric shock or burn.

Even when the MR system is switched off, mains voltage at the power input of the RFPA 35 kW (Astex/ MKS) is present.

- » Before any work on the amplifier switch off the circuit breakers CB2 and then CB1 at the amplifier and at least circuit breaker F25 in the mains box (CCA) and ensure against accidental switch-on.
- » Observe the warning information on the amplifier.
- » After the amplifier has been turned off, allow at least ten minutes for the high voltage filter capacitors to discharge fully before removing any of the cover plates.



Observe the “Unpacking and packing instructions” attached at the transport box of the Astex/ MKS RFPA.

1.1.2.1 RFPA (Ha/ Sy/ Sonata)

Assembly	Adjustment required	Final checks
RFPA (DORA/CORA)	Perform TuneUp: ■ RF CharacteristicBC/LC	Perform Quality Assurance: ■ CoilCheck(SN) ■ RF Verify BC/LC ■ Calc Artefacts
For replacement refer to (and following)		
Transmit cable	No adjustments required	Perform Quality Assurance: ■ CoilCheck(SN)

1.1.2.2 RFPA Trio

Assembly	Adjustment required	Final checks
RFPA 8kW	Perform TuneUp: ■ RF CharacteristicBC/LC	Perform Quality Assurance: ■ CoilCheck(SN) ■ RF Verify BC/LC ■ Calc Artefacts
For replacement of RFPA 8kW refer to (and following)		

Assembly	Adjustment required	Final checks
RFPA 35kW (ASTEX) PS Deck	■ Mains accommodation Perform TuneUp: ■ RF CharacteristicBC/LC	Perform Quality Assurance: ■ CoilCheck(SN) ■ RF Verify BC/LC ■ Calc Artefacts
RF Deck	Perform TuneUp: ■ RF Characteristic	
PA tube IPA tube		
CAN Bus Controller	No adjustments required	
For replacement refer to (and following)		
Transmit cable	No adjustments required	Perform Quality Assurance: ■ CoilCheck(SN)

1.1.3 RFAS

Assembly	Adjustment required	Final checks
LCCS D11	Perform TuneUp: ■ Tuning Calibration ■ Receive Path Calibration	Perform Quality Assurance: ■ Receive Path Cal Check
BCCS D5	Perform TuneUp: ■ Tuning Calibration	Perform Quality Assurance: ■ CoilCheck(SN)
TALES	Perform TuneUp: ■ Tuning Calibration ■ RF Characteristic ■ Coil Power Losses	Perform Quality Assurance: ■ Coil Check (SN_DIP) ■ Coil Power Losses Check
TAS	No adjustments required	Perform Quality Assurance: ■ RF Verify BC/LC
SAMI	Perform TuneUp: ■ RF CharacteristicLC	Perform Quality Assurance: ■ RF Verify LC
BTB	Perform TuneUp: ■ Tuning Calibration	Perform Quality Assurance: ■ CoilCheck(SN)
RF-cable between TALES and BTB	Perform TuneUp: ■ Tuning Calibration	Perform Quality Assurance: ■ CoilCheck(SN)

1.1.4 RFCI

**CAUTION**

There is the risk of irreversible destruction of the device by freezing up of the internal water pipes during shipping.

Before returning a replaced power supply unit of the RFCI, drain the water from the pipes of the device.

» Collect the water/cooling fluid and use it for refilling the cooling system



Jumper on motherboard D14, 57 73 085.

Setting A (upper row):

High field systems (>0.8T) as Rhapsody, Harmony Symphony, Sonata.

Trio: Setting A, except X60 -> setting B

Setting B (lower row):

Low field systems (<0.6T) as MAGNETOM Open, Concerto.

Assembly	Adjustment required	Final checks
PowerSupply	No adjustments required	Quality Assurance/ Coil-Check(SN) and Calc Artifacts
Motherboard D14, 57 73 085	Check jumpers! ■ Setting A: Highfield systems ■ Trio: Setting A; X60 setting B. ■ Setting B: Lowfield systems	Perform Quality Assurance: ■ CoilCheck(SN)
DYSCON	No adjustments required	Perform Quality Assurance: ■ CoilCheck(SN)
Service Board	No adjustments required	
CAN Module	No adjustments required	Perform Quality Assurance: ■ CoilCheck(SN)

1.1.5 PIN Booster (Trio)



CAUTION

The PIN Booster contains larger magnetic components such as mains transformer, fan with magnetic housing a.s.o

There is the risk of physical injury for the service engineer and damage to the components, if it is attracted by the 3T magnetic field.

The following precautions have to be observed:

- » Minimal distances between the PIN booster and magnet covers (magnetic field is on):
- » Lateral distance from cover (x-direction) > 0.5 m Axial distance from funnel opening (z-direction) > 1.5 m
- » If the PIN booster must be assembled at the RF-filter plate behind the magnet in distance less than 1.5 m, the magnet has to be ramped down

Assembly	Adjustment required	Final checks
PIN Booster	No adjustments required	Perform Quality Assurance: ■ CoilCheck(SN) for BC and a local coil.
For replacement refer to (→ Pin Booster (Trio) / Page 42).		

1.1.6 Coils

Assembly	Adjustment required	Final checks
Body Coil Ha/ Sy/ Sonata water cooled) (Refer to (→ Body Coil (Ha/ Sy/ Sonata) / Page 13))	Perform TuneUp: ■ Image Brightness ■ Coil Power Losses (→ Check the Adjustment of the Cooling Water Flow / Page 21))	Perform Quality Assurance: ■ CoilCheck(SN) ■ Coil Power Losses Check ■ Calc Artefacts ■ Spike Check
Water quick connectors (Refer to)	No adjustment required	Check tightness of quick connectors
Body Coil Type 2 (Refer to (→ Body Coil Type-2 / Page 30))	Perform TuneUp: ■ Body Coil Tuning ■ Coil Power Losses ■ Image Brightness (Expert Mode)	Perform Quality Assurance: ■ CoilCheck(SN) ■ Coil Power Losses Check ■ Calc Artefacts ■ Spike Check

Assembly	Adjustment required	Final checks
Body Coil Trio (BCT) (Refer to (→ Body Coil (Trio) / Page 36))	Perform TuneUp: ■ Body Coil Tuning ■ Coil Power Losses ■ Image Brightness	Perform Quality Assurance: ■ CoilCheck(SN) ■ Coil Power Losses Check ■ Calc Artefacts ■ Spike Check
Local coils	No adjustment required	Perform Quality Assurance: ■ CoilCheck(SN) ■ Calc Artefacts



CAUTION

During the working on the Body Coil or performing BC tuning (refer to (→ Body Coil Tuning (TuneUp) for Body Coil-2 and Body Coil Trio / Page 31)) you have to work close by the non-insolated connections of the gradient cables at the gradient coil.

Hazardous voltages may be present at the connections of the gradient cables !

- » Switch-off the gradient power amplifier (by circuit breaker F14/ GSSU in the CCA). Wait at least 20 minutes from this time before you start the BC Tuning adjustments.

First switch off the gradient power amplifier by F14 GSSU in the CCA.

To get access to the trimmer capacitors at the backside of the Body Coil you have to remove the back funnel of the magnet cover at the service end.

The gradient power amplifier must be switched off for at least 20 minutes before performing the BC Tuning procedure, i.e. adjustments of the trimmer capacitors at the body coil. In the meantime you should remove the funnel.

For turning the trimmer capacitors you need the special antimagnetic adjustment tool (i.e. long screwdriver), material number 73 66 771.

1.1.7 Overview Body Coil Types

System	Body Coil	Material number	Polarization	Water cooling	Foam cushions	RF cable between BTB and BC
Harmony	HF Body Coil 024	11 40 438	linear	yes	no	11 39 695
	HF Body Coil 024	59 37 946	linear	yes	no	11 39 695
Impact Harmony	HF-Body Coil 087/MU3 S	57 26 869	linear	yes	yes	59 37 904
	HF-Body Coil 087/MU3 L	57 26 869	linear	yes	yes	59 37 904
Symphony	HF-Body-Coil 047	11 40 446	circular	yes	no	11 39 703
	HF-Body-Coil 047 S	59 37 953	circular	yes	yes	11 39 703
	HF-Body-Coil 047 Q	59 41 203	circular	yes	yes	11 39 703
	HF-Body-Coil 047-2	73 66 607	circular	no	yes	11 39 703
Vision Symphony	HF-Body-Coil 089	59 39 819	circular	yes	yes	59 39 900
	HF-Body-Coil 089-2	73 66 631	circular	no	yes	59 39 900
Sonata	HF-Body-Coil Sonata	59 38 886	circular	yes	yes	1139 703
	HF-Body-Coil Sonata-2	73 66 623	circular	no	yes	1139 703
Vision Sonata	HF-Body-Coil 089 Sonata	59 42 771	circular	yes	yes	59 39 900
	HF-Body-Coil 089-2 Sonata	73 66 649	circular	no	yes	59 39 900
Trio (whole body)	HF-Body-Coil Trio (whole body)	73 66 441	circular	no	no	73 66 425



The red crossed out items are no more available as spare parts. For the Body-Coil xxx-2 the corresponding BTB047-2 is necessary.

Check the correct RF cable between BTB and BC-2, especially on upgrade systems. Otherwise problems can arise during TuneUp / BC Tuning.

2.1 Body Coil (Ha/ Sy/ Sonata)



CAUTION

The foamed plastics and the plastic film around the body resonator serve for noise protection (→ Fig. 1 Page 15).

- » Do not remove the cover.
- » Never deposit the body resonator on a sharp-edged surface. The plastic cover, damping material and the capacitors could be damaged.

2.1.1 Tools Required

- Manual vacuum pump material number 7057318, not delivered with the new Body Coil.
- Non-magnetic tool kit.
- Non-magnetic measuring tape or gauge.
- Non-magnetic spirit level.
- Special brass wrench, used to unscrew/tighten the RF transmit cable. This wrench is part of the system shipment.

2.1.2 Prerequisites

2 types of Body Coils are in the field. One with a mechanical stop and one with a positioning label (refer to (→ Fig. 7 Page 18)). Because the z-alignment depends on the used type, follow the instruction below before the Body Coil has been removed.

1. Defective coil with mechanical stop - new coil with position label:

Measure and note the distance X between the edge of the gradient coil and the edge of the Body Coil (refer to (→ Fig. 8 Page 19)). Measure and note the distance Y between the front edge of the body resonator assembly and the mechanical stop as shown in (→ Fig. 7 Page 18).

2. Defective and new coil with positioning label:

Measure and note the distance X between the edge of the gradient coil and the edge of Body Coil (refer to (→ Fig. 8 Page 19)). Measure and note the distance T between the front edge of the Body Coil and the label (arrow-head) as shown in (→ Fig. 9 Page 20).

3. Defective and new coil with mechanical stop:

Measure and note the distance X between the edge of the gradient coil and the edge of the Body Coil (refer to (→ Fig. 8 Page 19)).

A Body Coil Type 2 has to be installed mechanically as other new coils (with foam cushions). For differences in the installation procedure refer to (→ Body Coil Type-2 / Page 30).

2.1.3 Removal and Installation Procedure

- Switch-off the MR system.
- Loosen the lower bolt of the front funnel.
- Move the patient table down.
- Switch-off the MR system completely.
- Remove the display and the light marker.
- Remove the front funnel.
- Remove the back funnel, refer to (→ Magnet Installation and Cabeling / M1-010.812.01) (11 Allen bolts; 2 connectors of patient video monitoring camera).
- Remove the RF-transmit cable from the resonator using the special wrench supplied (1 cable for Harmony LP Body Coil, 2 cables for Symphony CP Body Coil).
- Remove the 4 8 mm Allen bolts at the back of Body Coil; don't loosen the nut and bolts of the suspension. Refer to the example in (→ Fig. 2 Page 15) and (→ Fig. 3 Page 15) for Harmony.
- Remove the 2 quick connect water lines at the back of the Body Coil (refer to the example in (→ Fig. 3 Page 15) for Harmony).
- Open the lid (one for Harmony, 2 lids for Symphony) inside the Body Coil and remove the ground plug (refer to (→ Fig. 2 Page 15) for Harmony).

NOTICE

This bolt does NOT tighten the ground connection. The ground connection is simply plugged in by hand

» **Do not loosen or tighten the 10-mm bolt at the ground plug.**

- Remove the 4 8 mm Allen bolts and the 2 quick connect water lines at front of the Body Coil.
- Evacuate the noise damping cushions (if installed) with the help of the manual vacuum pump or the electrical vacuum pump (comfort kit), if installed. Close the hoses with cable ties or plug and pull out the cushions. Refer to (→ Fig. 1 Page 15) and (→ Fig. 4 Page 15).
- Slowly and carefully pull the Body Coil OUT of the bore (patient table side), pay attention to the ground connectors and block capacitors.



Do not touch the resonator directly above the RF-connectors when pulling it out. These areas are extremely fragile.

Fig. 1: Manual vacuum pump and damping cushions

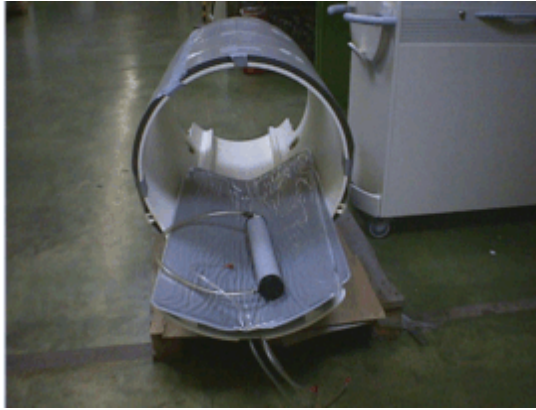
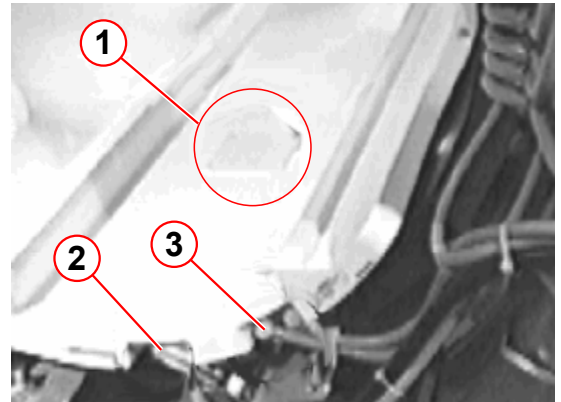
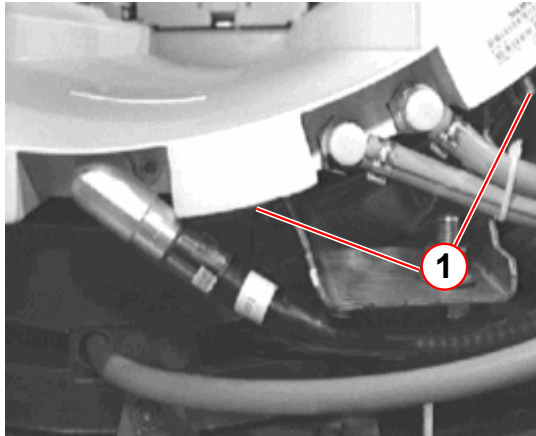


Fig. 2: RF-connector / quick connect water lines (Harmony)



- (1) lid of ground plug
- (2) RF
- (3) water

Fig. 3: Suspension of the body coil (Harmony)



- (1) 8 mm Allen bolts
- (2) 8 mm Allen bolts

Fig. 4: Damping cushions partly removed (Symphony)



- For reinstallation proceed in reverse order, but observe the following advice:
 The foamed plastics and the plastic film around the body resonator serve for noise protection. Do not remove that cover. **Before inserting the Body Coil, cover the front suspensions (e.g. with a towel or cardboard). This should protect the plastic cover and damping material around the Body Coil when the coil is inserted.**
 Two persons should hold and push the coil in front of the magnet; the second person should lift the coil at back of the magnet when the coil reaches its end position.
Pay attention to the block capacitors at the ground connectors of the RF-shield during the coil slides in.



Take care of the water connections at the bottom of the Body Coil while positioning the coil on the back suspensions.

The coil must be lifted carefully a little bit.

The water hoses with their quick connectors must be connected to the Body Coil plugs free of stress.

This will avoid damage to the connections and successive water leakages.

- Reinstall the evacuated damping cushions. Attach the cushions with the cutout at the bottom, the other 2 at the top of the Body Coil.
Refer to (→ Fig. 4 Page 15). The cushions should be vented after alignment of the coil.
- Refer to the following instructions for aligning the body resonator.

2.1.4 Alignment of the Body Coil

After installing a new body resonator, check the alignment of the Body Coil to the gradient coil and adjust it, if necessary.

2.1.4.1 Case 1: The Body Coil rests smoothly on the Supports.

- Measure Z-distance and radial distances (front and back), refer to (→ Z-direction check/adjustment / Page 18) and (→ Fig. 5 Page 17); use a non-magnetic spirit level to check the horizontal position.
- If the position and distances are ok, then fix the Body Coil with the mounting bolts on the supports (torque = 15 Nm). Use new serrated lock washers.
- Secure the mounting bolts with Loctite 221.
- Vent the vacuum cushions
- Check the radial distances (→ Fig. 6 Page 18); if not ok -> Case 2
- Complete the assembly and finally measure the decoupling (refer to Update Instruction MR 009/99/1, Body Coil decoupling check).

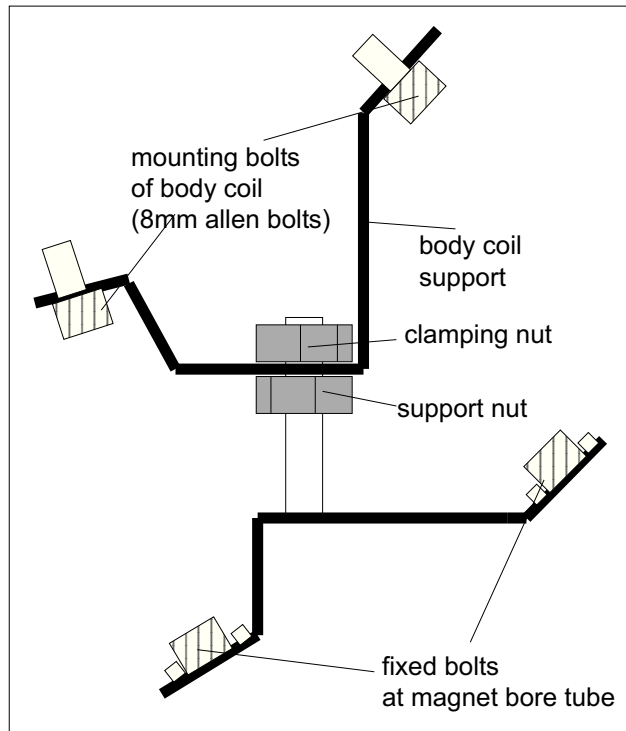
2.1.4.2 Case 2: The Body Coil doesn't rest smoothly on the Supports

(The distances may be out of tolerance)

- Loosen the clamping nuts (→ Fig. 5 Page 17).
- Fix the Body Coil with the mounting bolts on the supports (torque = 15 Nm). Use new serrated lock washers.
- Secure the mounting bolts with Loctite 221.
- Measure Z-distance and adjust it if necessary; refer to (→ Z-direction check/adjustment / Page 18).
- Vent the vacuum cushions.
- Measure the radial position (front and back) and adjust with the support nuts (→ Fig. 5 Page 17) if necessary (refer to (→ Fig. 6 Page 18)). The Body Coil must rest on the supports without bracing.
- Tighten the clamping nuts.

- Check radial distances (refer to (→ Fig. 6 Page 18)).
- Complete the assembly and on Symphony systems finally **measure the decoupling, refer to Update Instruction MR 009/99/1, Body Coil decoupling check**
 - » The spec for the transmission value is: Transmission < -14 dB; after proper adjustment of BC it should be lower than -20 dB.

Fig. 5: Sketch of body coil support

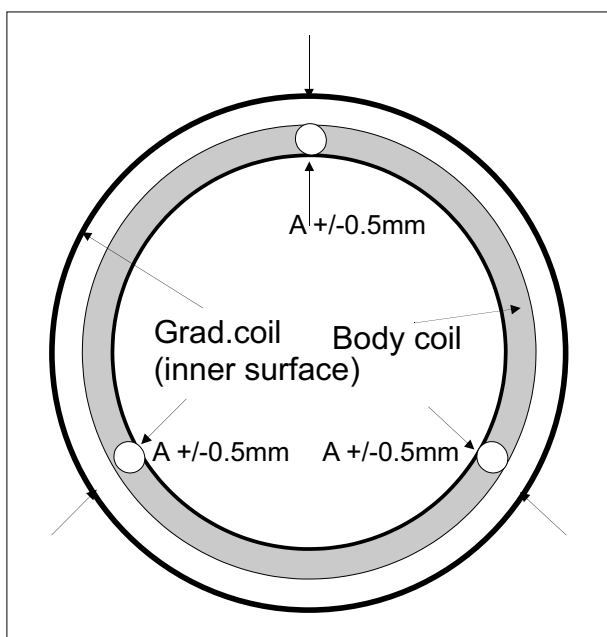


2.1.4.3 Centric Alignment of BC to the Gradient coil/horizontal Position

- Measure the distance between the inner surface of the gradient coil (RF-screen) to the inner surface of the Body Coil tube (front and back) at 3 points shifted by 120 degrees. One of these points should be at the top of the Body Coil. Use a non magnetic gauge.

- Adjust the Body Coil suspension that all measured distances are within a range of 1 mm.

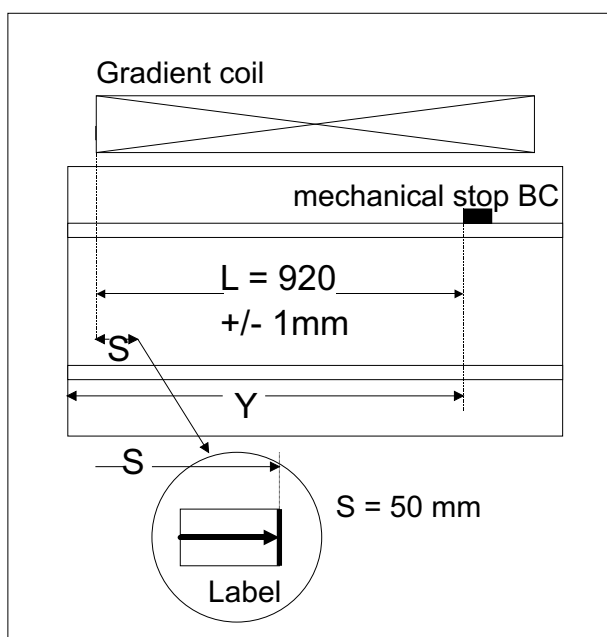
Fig. 6: Centric alignment to the gradient coil



- Leveling with non-magnetic spirit level:
With the level placed crossways on the patient table guide rails, measure and adjust the horizontal alignment of Body Coil.

2.1.4.4 Z-direction check/adjustment

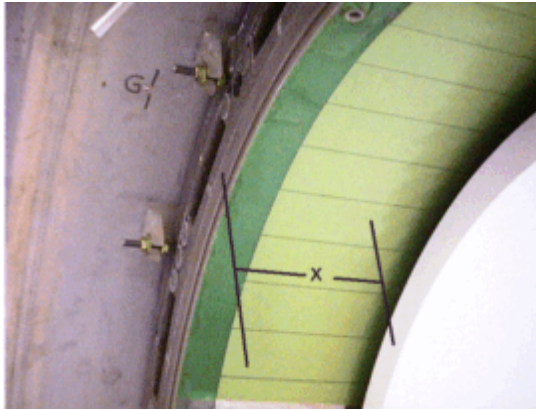
Fig. 7: Z-direction alignment of body resonator



■ **Defective coil with mechanical stop-new coil with position label:**

- Before the defective Body Coil was removed, the distance **X** as shown in and the distance **Y** as shown in (→ Fig. 8 Page 19) has been measured.
- At the new Body Coil attach a temporary marking at the same distance **Y**.

Fig. 8: Distance **X** between edges of gradient coil and BC



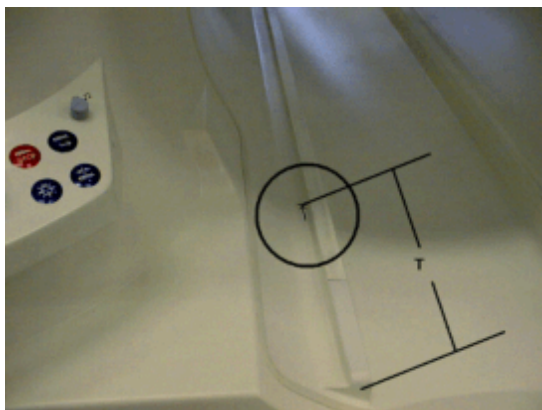
- After installation of the new Body Coil measure again the distance **X** between the edges of gradient- and Body Coil and the distance **L** = 920 mm between the temporary marking and the front edge of the gradient coil. If these distances differ more than ± 1 mm from the original distances, adjust the suspension of the Body Coil (refer to (→ Case 2: The Body Coil doesn't rest smoothly on the Supports / Page 16)).
- Finally attach the label for positioning the shim device beside the left patient table guide rail at a distance **S** = 50 mm (arrow-head) from the front edge of the gradient coil (refer to (→ Fig. 7 Page 18) and (→ Fig. 8 Page 19)).

■ **Defective and new coil with positioning label:**

- Before the defective Body Coil was removed, the distance **X** and the distance **T** as shown in (→ Fig. 9 Page 20) has been measured.
- After installation of the new Body Coil measure again the distance **X** between the edges of gradient- and Body Coil.
If this distance differs more than ± 1 mm from the original distance, adjust the suspension of the Body Coil (refer to (→ Case 2: The Body Coil doesn't rest smoothly on the Supports / Page 16)).

- Finally attach the label for positioning the shim device beside the left patient table guide rail at distance **T** from arrow-head to the front edge of the Body Coil assembly.

Fig. 9: Distance **T** between front edge of guide rail and arrow head of label



The distance between the front edge of the gradient coil and the arrow head of the label must be $S = 50 \text{ mm}$ (refer to (→ Fig. 7 Page 18)).

■ **Defective and new coil with mechanical stop:**

- After installation of the new Body Coil measure again the distance **X** between the edges of gradient- and Body Coil and the distance **L** = 920 mm between the mechanical stop and the front edge of the gradient coil (smooth offset ring), refer to (→ Fig. 7 Page 18). If these distances differ more than $\pm 1 \text{ mm}$ from the original distances, adjust the suspension of the Body Coil (refer to (→ Case 2: The Body Coil doesn't rest smoothly on the Supports / Page 16)).

Only for information:

The distance **X** depends on a measure "**M**", which is printed on the magnet at patient side in the following way:

M	X (+/- 1 mm)
5	163
4	163
3	163
2	162
1	161
0	160
-1	159
-2	158
-3	157

M	X (+/- 1 mm)
-4	157
-5	157

2.1.5 Final Checks

2.1.5.1 Check the Adjustment of the Cooling Water Flow

- Check the correct installation of the flow sensor. The arrow on the sensor must point at flow direction (out of BC).



In case of sporadic error messages "Body Coil cooling error" it is recommended to open the flow reducer (refer to (→ Fig. 10 Page 22)) completely (turn counterclockwise).

- Connect the adapter cable (→ 2/ Fig. 11 Page 22) into the flow sensor circuit. Refer to (→ Fig. 10 Page 22) and (→ Fig. 11 Page 22).
- Set the Fluke 8060 DVM to AC in the 2V range and connect it to the adapter connector (→ 1/ Fig. 11 Page 22), using long measuring leads.



WARNING

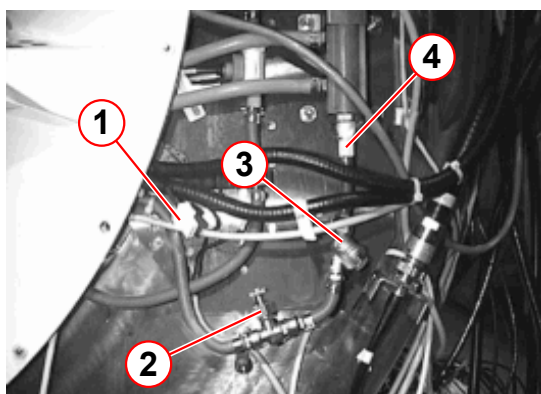
The Fluke DVM could be damaged if it is used too close (distance < 1m) to the energized magnet.

- » The Fluke 8060 DVM has magnetic objects, keep distance from magnet!

- Set the Fluke to "Hz" to measure the frequency of the voltage.
- The frequency should be set to the frequencies listed in the tables below, otherwise adjust the reduction valve (→ 2/ Fig. 10 Page 22). The serial number of the Body Coil is attached at the rear side of the coil.

Serial no.of Body Coil	Revision level ES	Material number	Frequency value
≤ 1026 (Harmony)	≤ 02	1140438 K2204	40 Hz ± 3 Hz
> 1026 (Harmony)	> 02	1140438 or 5937946 K2204	70 Hz ± 3 Hz
≤ 1009 (Symphony)	≤ 02	1140446 K2214	40 Hz ± 3 Hz
> 1009 (Symphony)	> 02	1140446 or 5937953 K2214	70 Hz ± 3 Hz

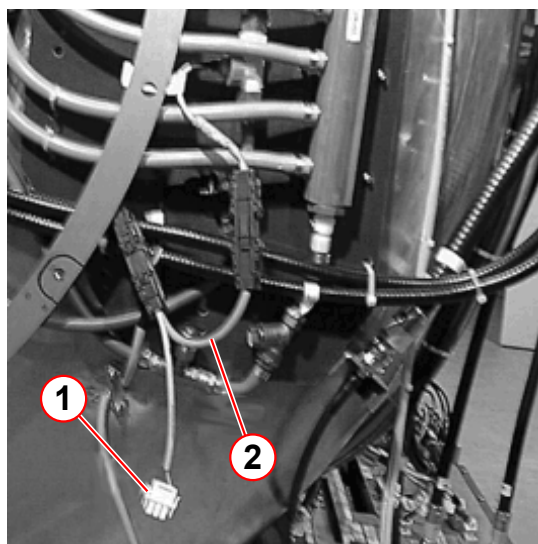
Fig. 10: Adjustment check of cooling water flow



- (1) Adapter connector
- (2) Reduction valve
- (3) Small water filter
- (4) Quick connector

- If the nominal value cannot be obtained, clean the small water filter (→ 3/Fig. 10 Page 22) in front of the reduction valve as follows.
 - Disconnect the two quick connectors (→ 4/Fig. 10 Page 22) of the Body Coil cooling circuit at the gradient coil water distributors.
 - Open and clean the small water filter (→ 3/Fig. 10 Page 22).
 - Close the filter again and reconnect the 2 quick connectors.
 - Check the correct installation of the flow sensor. The arrow on the sensor must point at flow direction (out of BC).
 - Repeat the adjustment.
 - Check for water leakages.
 - Check the gradient coil flow with the Software (30 -32 l/ min.).

Fig. 11: Adapter cable /connector for frequency measurement



- (1) Adapter connector
- (2) Adapter cable



After the adjustment check, remove the adapter cable from the flow sensor circuit to prevent spikes.



The monitoring of the cooling water flow and temperature (return flow) is done at supervision board D9 K2209, which is located in the patient table electronics.

- Perform TuneUp Expert Mode / Coil Power Losses and Image Brightness.
- Perform Quality Assurance / Coil Check (SN), Coil Power Losses Check , Calc Artefacts and Spike Check .

2.1.6 Shipment of the Body Coil



CAUTION

There is the risk of irreversible destruction of the device by freezing up of the internal water pipes during shipping.

- » Before returning a replaced water cooled Body Coil, power supply unit of the RFCI or the RFPA, drain the water from the pipes of the device by opening both ends.

- Drain the water from the cooling pipes on both sides of the resonator with the help of the enclosed quick connectors.

2.1.7 Repairing RF Body Coils and Water Manifolds with Broken-off Plastic Quick Disconnectors

2.1.7.1 Explanation

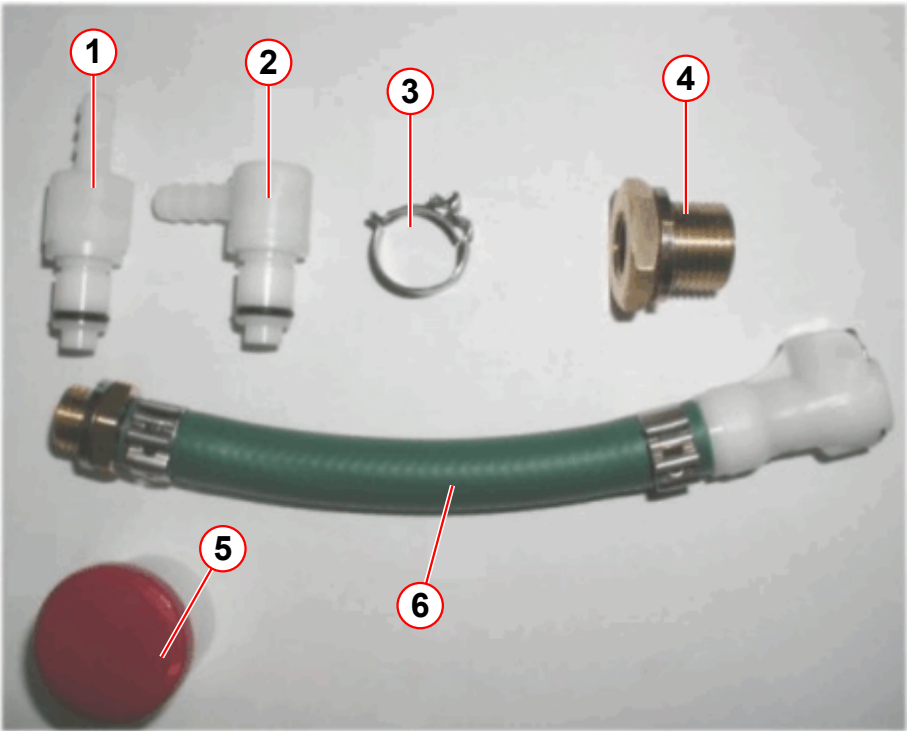
We had a number of Body Coils with leaking or sheared-off quick disconnectors as well as damaged water manifolds in the field. This causes system downtime as well as high repair costs. To decrease costs and downtime, it was suggested to repair the damage in the field by replacing the quick disconnectors on site.

Tools required (all non-magnetic)

- 17/19 mm spanner
- Heat gun (adjustable)
- large flat blade screwdriver (non-magnetic)
- self-made wire brush
- quick disconnector repair kit (material number1 141758)

Contents of Repair Kit Material Number 1141758

Fig. 12: Repair kit



- (1) Quick disconnect, straight
- (2) Quick disconnect, 90° elbow
- (3) Hose clamp
- (4) Adapter 3/4 " to 1/2 "
- (5) Water sealer
- (6) Water hoses, tailored

Tab. 1

Item	Amount	Description
1	2	Quick disconnect, straight
2	2	Quick disconnect, 90° elbow
3	3	Hose clamp
4	1	Adapter 3/4 " to 1/2 "
5	1	Water sealer
6	4	Water hoses, tailored

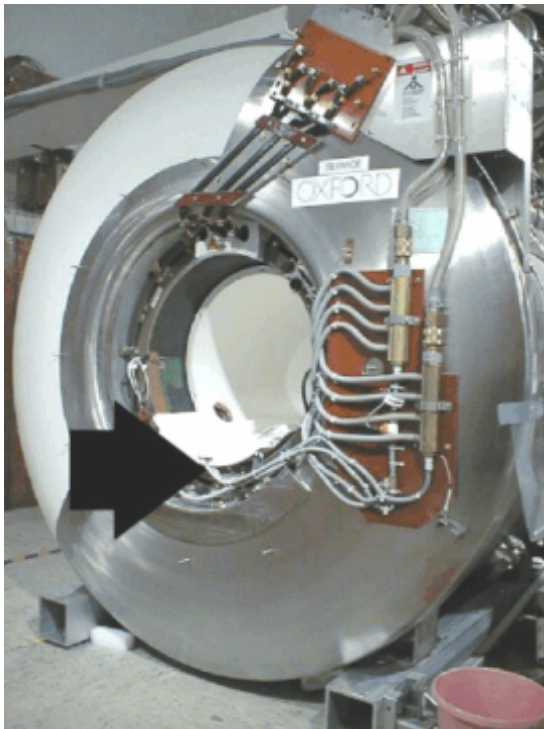
Body Coil Repair

Fig. 13: Self-made brush



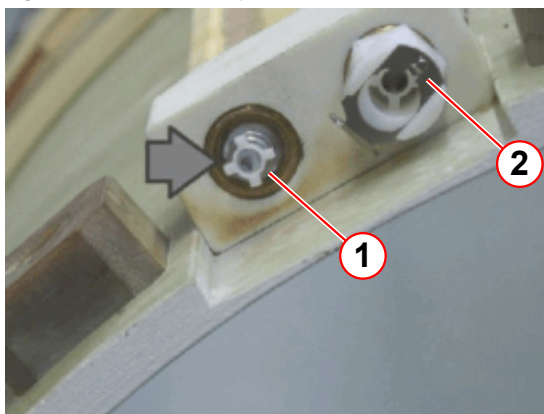
- Prepare a 10 cm long piece of 3 or 4 gauge flexible wire, strip off 1 cm of insulation, and bend the wire strands as shown in (→ Fig. 13 Page 25).

Fig. 14: Position of quick disconnects in the rear of the magnet (see arrow)



The quick disconnects are located in the front and back of the magnet.

Fig. 15: Sheared-off quick disconnecter (see arrow)



- (1) spring and plastic piece
- (2) break this off

When a quick disconnecter leaks due to a hairline crack, break off the piece that protrudes from the front of the Body Coil (→ Fig. 15 Page 26).

- The remaining pieces of the female quick disconnecter, a short plastic stump and spring, have to be removed completely.

Fig. 16: Sheared off quick disconnecter



- The stump has to be removed first. Start by removing the spring and the plastic piece (→ Fig. 16 Page 26) using flat nose pliers

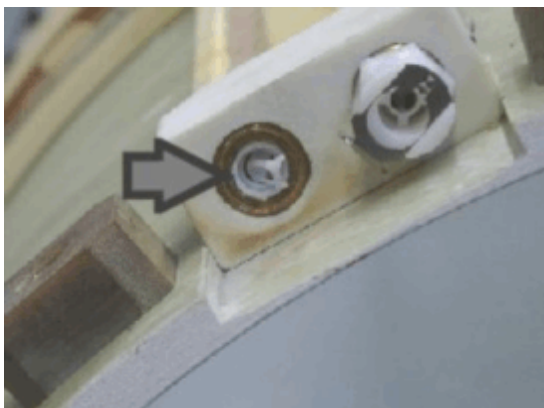


CAUTION

The spring is magnetic - do not drop it.

- » It may be drawn into the magnet

Fig. 17: Plastic stump remains in the bushing



- (→ Fig. 17 Page 26) shows the spring removed

Fig. 18:



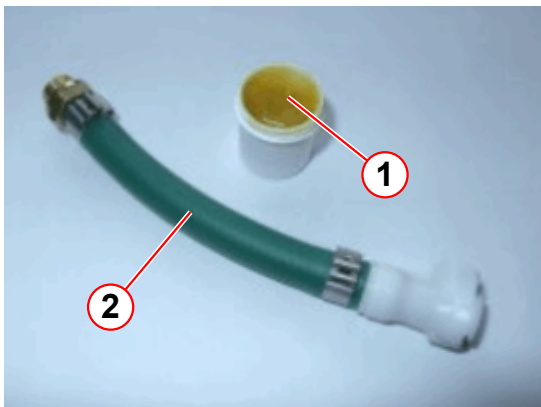
- Use the largest flat blade screwdriver from the non-magnetic tool kit.
- Insert as shown in (→ Fig. 18 Page 27) and use sufficient force to turn.
- If it cannot be moved, try using a hot air fan to heat up the sealing adhesive applied at the factory.
- Turn ccw and completely remove the stump.

Fig. 19: Clean the threads



- Thoroughly clean the threads using the spiral brush (→ Fig. 19 Page 27).

Fig. 20: New quick disconnect with water hose and sealer



- (1) Water sealer
(2) Water hoses, tailored

The repair kit contains the necessary items water hose plus sealer (see (→ Fig. 20 Page 27))

Fig. 21: Apply sealant



- Take one of the green hoses from the repair kit and apply 1 layer of sealer to the brass fitting (→ Fig. 21 Page 28).

Fig. 22: Quick disconnects after repair



(1) O-ring

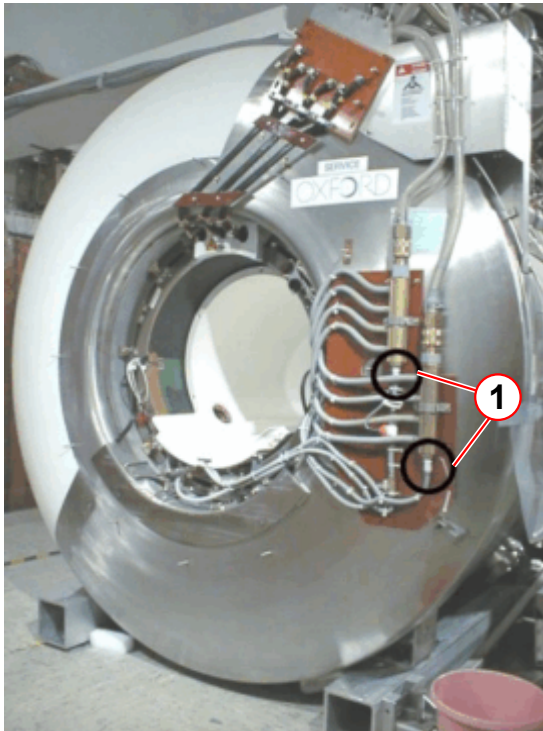
- Screw into the brass bushing and tighten by hand (→ Fig. 22 Page 28).



It should be sufficient to tighten the new hose by hand until the green hose starts slipping on the brass nipple. Then immediately stop turning it any further

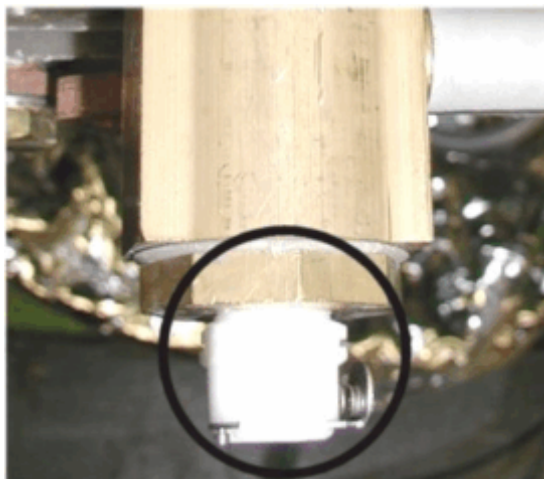
2.1.8 Repairing the Quick Disconnects at the Main Water Manifolds

Fig. 23: Location of quick disconnects encircled



(1) Quick isconnectors

Fig. 24: Quick disconnecter at main manifold

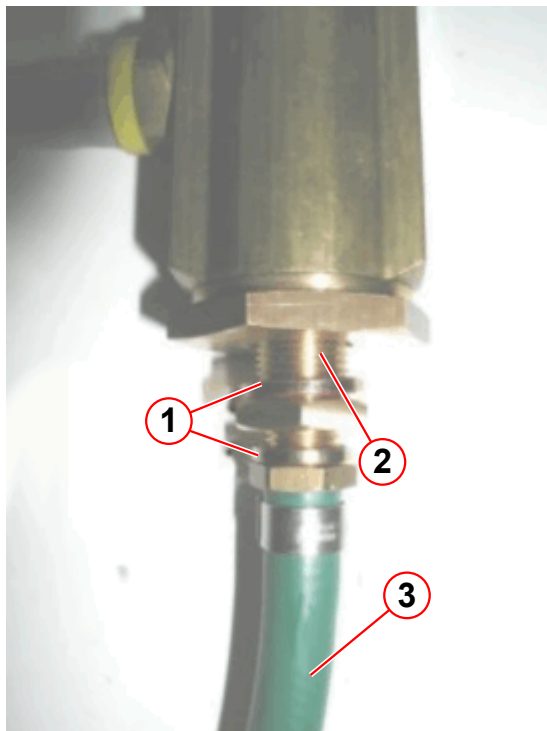


If a quick disconnecter breaks off or leaks at one of the main water manifolds in back of the magnet (→ Fig. 24 Page 29) use the very same procedure as described earlier to rectify the problem.

- Remove the white plastic female quick connector.
- Take one green hose with new fitting.
- Apply sealer to the adapter and hose.
- Screw adapter into manifold and tighten with non-magnetic open-end wrench.

- Screw hose into adapter and tighten with non-magnetic open-end wrench.

Fig. 25: Quick disconnectors after repair



- (1) O-ring
- (2) Adapter 3/4" to 1/2"
- (3) Water hoses, tailored

2.1.9 Body Coil Type-2

- The BC-2 requires a new BTB (BTB 049-2).
- The new BTB 049-2 has to be configured in the Service Software..
- A new coil file is available for the BC-2 due to its lower power losses. The Coil Power Losses are measured using TuneUp and QA procedures.
- The BC-2 does not require water cooling. The existing water hoses are not used when replacing an older BC type. The hoses will be shut off by the existing quick coupling valves.

On D9 (patient table electronics) the red monitoring LED's V83 left (water flow monitoring) and V63 right (return temperature of cooling water) will light up.

In addition, the green LED V83 right (Body Coil ok) will be off.

The error will be masked in the BTB047-2. The FOC between D9/U2 and BTB/U5 is no longer necessary.

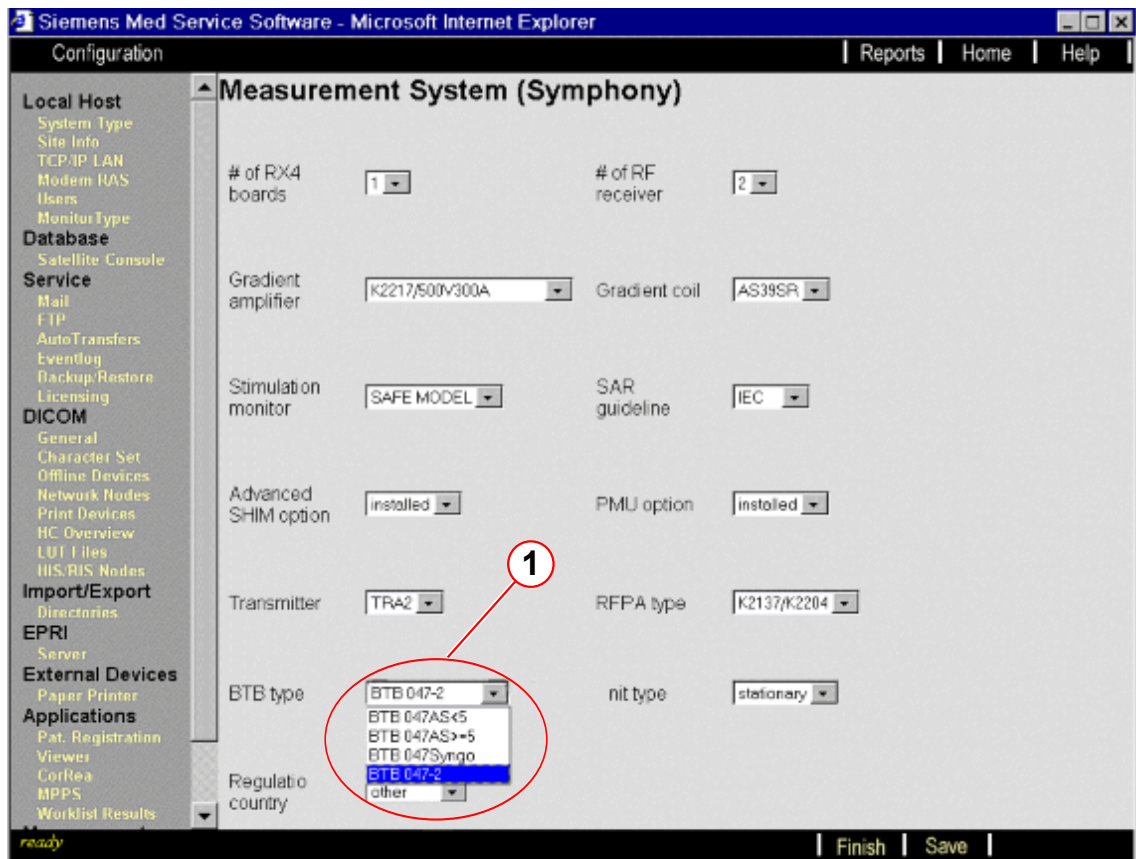
- "Body Coil Tuning" will be performed during TuneUp. Adjustments are normally not required. For replacement BC's it may be necessary to adjust the decoupling via trimmer capacitors (refer to (→ Body Coil Tuning (TuneUp) for Body Coil-2 and Body Coil Trio / Page 31)).
- Perform TuneUp / Coil Power Losses and Image Brightness (Expert mode).

- Perform Quality Assurance / CoilCheck(SN), Coil Power Losses Check , CalcArtefacts and Spike Check.

2.1.9.1 SW Configuration of BTB049-2

(Refer to (→ Software Installation syngo MR A30 (NUMARIS/4VA30A) / MR-000.816.26)).

Fig. 26: Configuration Platform - Measurement System



(1) BTB Type: BTB 047-2

- Select the following configuration in the "measurement" configuration platform:
- BTB Type: BTB 047-2
- Click "Save"
- Reboot the system according to the instructions in the configuration platform.

2.1.9.2 Body Coil Tuning (TuneUp) for Body Coil-2 and Body Coil Trio

Prerequisites

Since BC Tuning uses the H and S matrices to perform calculations, the Tuning Calibration TuneUp procedure has to be completed successfully before starting BC Tuning.

**CAUTION**

Hazardous voltages may be present at the connections of the gradient cables

During performing BC tuning you have to work close by the non-insolated connections of the gradient cables at the gradient coil

- » Switch-off the gradient power amplifier (by circuit breaker F14/ GSSU in the CCA). Wait at least 20 minutes from this time before you start the BC Tuning adjustments.

First switch-off the gradient power amplifier by F14 GSSU in the CCA.

To get access to the trimmer capacitors at the backside of the Body Coil you have to remove the back funnel of the magnet cover at the service end.

The gradient power amplifier must be switched off for at least 20 minutes before performing the BC Tuning procedure, i.e. adjustments of the trimmer capacitors at the Body Coil. In the meantime you should remove the funnel.

For turning the trimmer capacitors you need the special antimagnetic adjustment tool (i.e. long screwdriver), material number 73 66 771.

Fig. 27: Tool for trimmer capacitor adjustment of BC-2



Frequency and Decoupling Adjustment

You will be guided through making iterative fine adjustments. Between the adjustment steps, the resonance and decoupling of both systems (0 and 90 degree) can be varied manually using the 3 trimmer capacitors at the coil. The program stops when the resonance and decoupling are within specifications.

Fig. 28:

#21901 TuneUp
BCTuning 2002-01-17 180710

[Log File](#)

Please move the patient table out of the magnet.

Click OK to start the first measurement.

If the results are not in specification adjust the trimmer settings (see Online Help for details) and repeat the measurement by pressing the START button at the patient table control panel.

Reflection:

Transmission:

Step	Reflection (Minimum)					Transmission (Maximum)	
	Frequency		Delta	Factor		Frequency	Decoupling
	0 degree	90 degree		0 degree	90 degree		
	63.45 - 63.75 MHz	63.45 - 63.75 MHz		<100.0 kHz	0.0 - 0.3		
1	63.60	63.60	5.00	0.05	0.12	63.54	-14.25

The trimmer capacitors T1 and T2 affect the resonance frequencies of systems 1 and 2 and the capacitor TD affects the decoupling between the 2 systems.

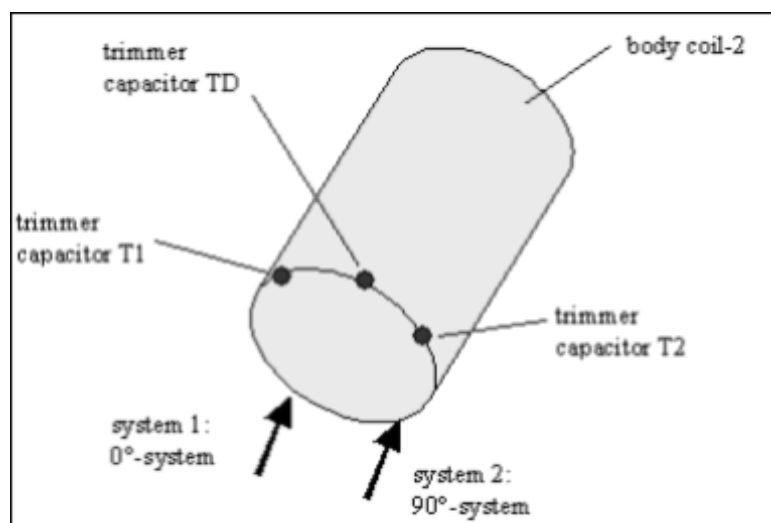
The decoupling has to be adjusted only if the measurement of the transmission value is not within specification.

The trimmer capacitors allow 50 turns. By default, they are set in the middle position which means you can make 25 turns clockwise or 25 turns counter-clockwise.



To prevent damage to the trimmers, use caution when approaching the end position. Do not use excessive force.

Fig. 29: Position of trimmer capacitors on BC-2



Trimmer capacitor	Main influence on	Clockwise (cw)	Counterclockwise (ccw)
T1	resonance frequency system 1	decrease resonance frequency 1	increase resonance frequency 1
T2	resonance frequency system 2	decrease resonance frequency 2	increase resonance frequency 2
TD	Decoupling between systems 1 and 2	increase capacitance	decrease capacitance



If the values are slightly out of spec, the system will still work.

Step 1	Reflection adjustment of delta frequency		
	delta < 100 kHz		
	yes	no	
	no action required	Frequency 0 degree > frequency 90 degree	
		yes	no
		turn T1 2x cw turn T2 2x ccw	turn T1 2x ccw turn T2 2x cw
		goto step 1	

Step 2	Mechanical adjustment of delta frequency		
	delta < 100 kHz ?		
	Transmission 0.. -12 dB		
	yes	no	
	Adjust the mechanical position of the BC in the gradient coil.	goto Step 3	
	goto step 2		

Step 3	Electrical adjustment of decoupling (transmission)		
	Transmission T = -12-dB .. -16 dB		
	yes	no	

Step 3	Electrical adjustment of decoupling (transmission)	
Step 3.1	Turn TD 3x cw, press start button at panel	
	T lower	
	yes	no
Step 3.1.1	Turn TD 3x cw, press start button at panel	Turn TD 3x ccw, press start button at panel
	T lower -> goto 3.1.1	T lower -> got 3.1.1
	no -> undo last step	no -> undo last step
Step 3.2	Turn T1, T2 2x cw, press start button at panel	
	T lower	
	yes	no
Step 3.2.1	Turn T1, T2 2x cw, press start button at panel	Turn T1, T2 2x ccw, press start button at panel
	T lower -> goto 3.2.1	T lower -> got 3.2.1
	no -> undo last step	no -> undo last step
If necessary, repeat Step 3.1		

no action required

2.2 Body Coil (Trio)

2.2.1 General

The Body Coil Trio is similar to the Body Coil type 2. Also the replacement procedure is almost the same as of the other Body Coils. The particularities are described in the following.

Without lifting device or special mounting tool the replacement of the BCT (Body Coil Trio) requires 4 persons for removing/inserting.

2.2.1.1 Physical Differences

- The BCT is longer and heavier as the BC2.
- There are three ground connections to the RF-shield at the gradient coil.
- The BCT has an additional connector to the static detuning unit (500 VDC).
- The BCT is assembled without vacuum cushions between BCT and gradient coil.

2.2.1.2 Measurements before Removing the old BCT

- Measure and note the distance "X" between edge of the BCT and gradient coil on the patient side. "X" should be 62.5 mm +/- 3 mm, depending of the position of the magnetic center, (refer to (→ Fig. 30 Page 36) and (→ Fig. 32 Page 37)).
Use a straight edge (or similar) across the front of the BCT to measure this distance precisely.
- Measure and note the distance "T" between the front edge of the guide rail in the BCT and the arrow head of the label for the shim device. "T" should be 112.5 mm +/- 3 mm, (refer to (→ Fig. 31 Page 36)).

Fig. 30: Distance X between edge of BCT and gradient coil

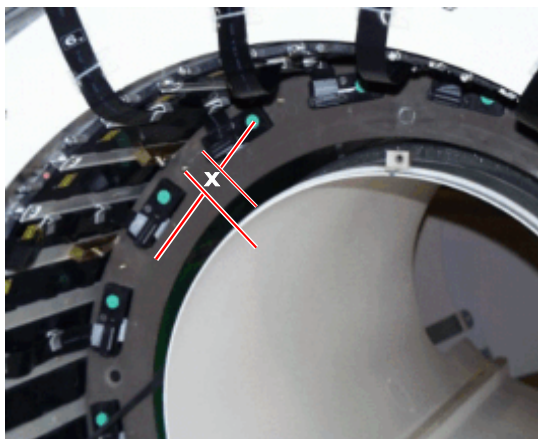


Fig. 31: Distance T between front edge of guide rail and arrow head of label

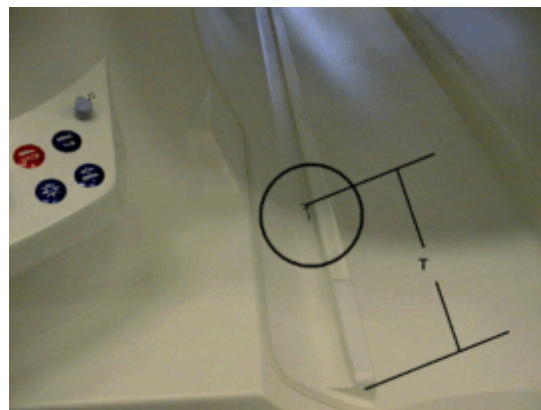
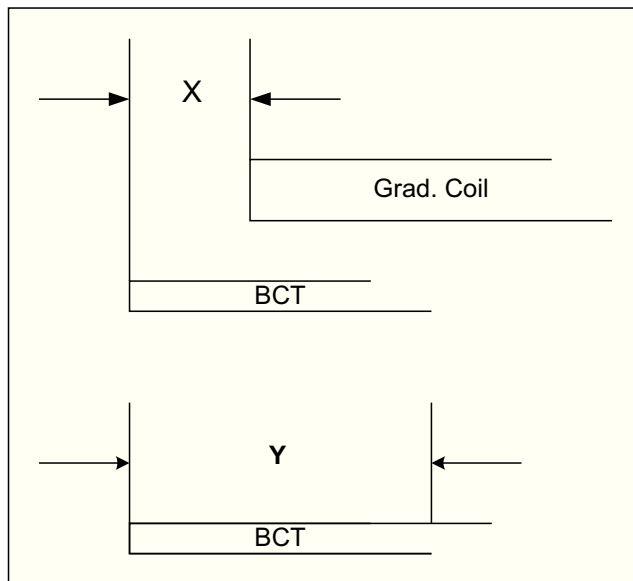


Fig. 32: Distances X and Y



Only for Information

The distances

X (= between edge of the BCT and gradient coil on the patient side) and

Y (= distance between front edge of BCT and the arrow head of the label for the shim device) - refer to (→ Fig. 32 Page 37) -

depend on a measure "**M**", which is printed on the magnet at patient side in the following way::

M	X (+/- 1 mm)	Y (+/- 1 mm)
5	59.5	109.5
4	59.5	109.5
3	59.5	109.5
2	60.5	110.5
1	61.5	111.5
0	62.5	112.5
-1	63.5	113.5
-2	64.5	114.5
-3	65.5	115.5
-4	65.5	115.5
-5	65.5	115.5

Positive values of M: Gradient coil is shifted to service end
Negative values of M: Gradient coil is shifted to patient end
M=0: Gradient coil center in magnetic center

2.2.1.3 Dis-/Reassembling the BCT

- Disconnect the RF-cables, the connector X3 from the BCT and the 3 ground connections to the RF-shield on the gradient coil.
- Remove the covers from both sides of the magnet.
- Remove the funnels at the front and the rear end of the magnet.
- Remove the top and bottom covers at the rear end of the magnet.
- At the rear and front end of the old BCT remove the 4 bolts attaching the coil to the support brackets.
- Adjust the table to a suitable height (hand crank).
- Slowly pull the old BCT out of the magnet.

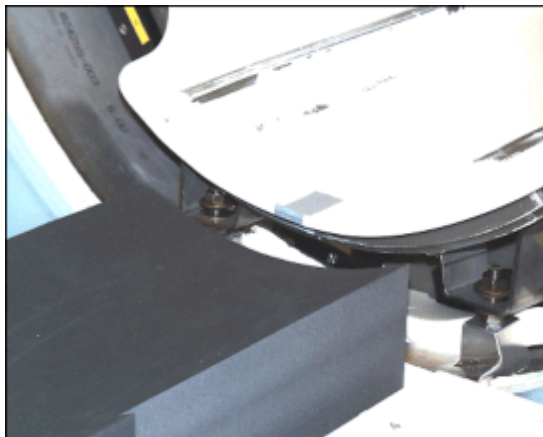
NOTICE

Use extreme care when removing the old BCT to avoid damage to the Faraday Shield inside the gradient coil.

Keep the coil centered at all times.

» **Particularly observe the capacitors soldered to the Faraday Shield**

Fig. 33: Service support on the patient table



- Let the BCT rest on the service support. During moving the BCT out, shift the service support accordingly, so that finally the center of the BCT rests on the center of the service support.
- Once the BCT is pulled out completely, lift it off the service support and store it in a suitable place on the floor.
- Place the new BCT on the service support on the patient table.
- Move the BCT into the gradient coil. 2 blocks mounted to the bottom of the BCT will slide on the 2 rails inside the gradient coil.

NOTICE

Use extreme care when moving the Body Coil into the magnet to avoid damage to the Faraday Shield inside the gradient coil.

Keep the coil centered at all times.

- » Particularly observe the RF-connectors at the rear side of the coil. They just pass clear of the capacitors soldered to the Faraday Shield.

- Place the BCT on the coil supports and insert all screws handtight.
- Connect the 3 grounding connectors (refer to (→ Fig. 34 Page 39), (→ Fig. 35 Page 39), (→ Fig. 36 Page 39), and (→ Fig. 37 Page 39)).

Fig. 34: Lower ground connection on RF-shield

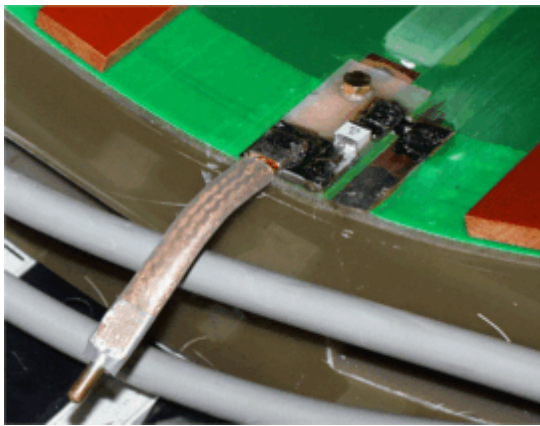


Fig. 35: Ground connection plug on bottom side of BCT

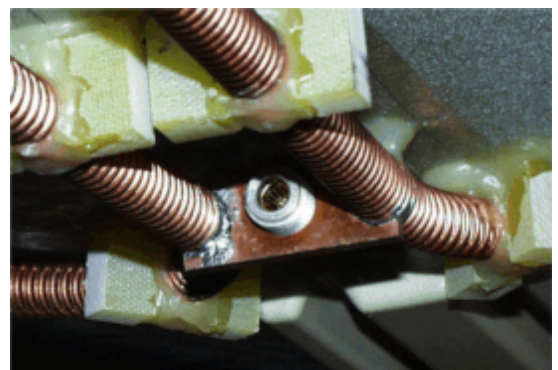


Fig. 36: Ground connection (right side) - same as on the left side

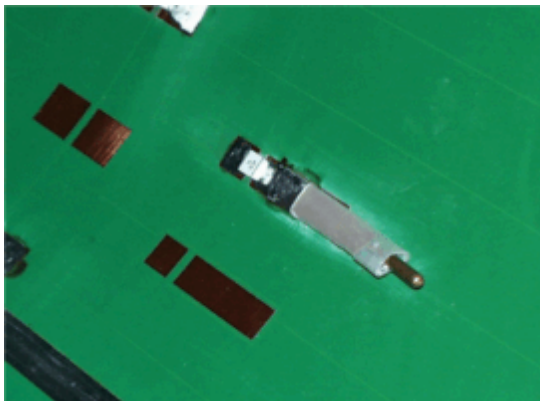


Fig. 37: Openings for access to left and right ground connections



- Adjustment in Z-direction at the same distance "X" as measured previously on the old BCT and gradient coil.

- Adjustment in x and y-direction:

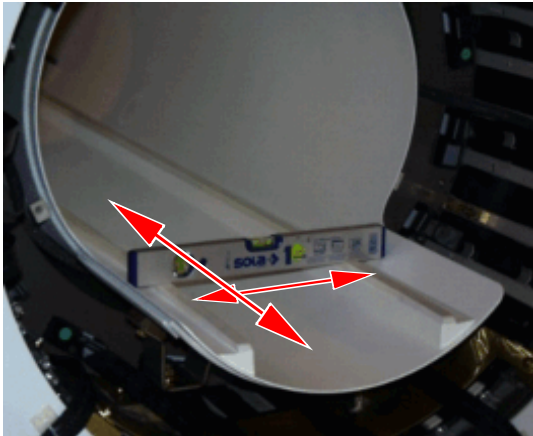
The Body Coil must be centered.

The distance between the Body Coil and the gradient coil must be the same all around (tolerance ± 0.5 mm).

Hint:

Use the clearance of a 19-mm non-magnetic open-end wrench as a gauge.

Fig. 38: Horizontal adjustment in x and z-direction



- Use an antimagnetic spirit level to adjust the Body Coil horizontally in x and Z-direction (refer to (→ Fig. 38 Page 40)).

- Tighten all screws and nuts of the coil supports.
- Attach the new label for the shim device position at previously measured position on "T" (arrow head) to the new BCT (refer to (→ Fig. 31 Page 36)).
- Reconnect the RF-cables and connector X3.
- For replacement BC's it may be necessary to adjust the decoupling via trimmer capacitors (refer to (→ Body Coil Tuning (TuneUp) for Body Coil-2 and Body Coil Trio / Page 31)).
- Perform TuneUp / Coil Power Losses and Image Brightness.

- Perform Quality Assurance: CoilCheck(SN), Coil Power Losses Check, Calc Artefacts and Spike Check.

Fig. 39: Connector X3 at static detuning unit at BC



Check of the PIN-diodes of Body Coil Trio:

For a quick check of the PIN-diodes remove the connector X3 of the detuning unit at the BC and measure with the "diode test function" of a DVM in both (plus/minus) directions at the detuning unit.

Typical displayed values: forward direction: 0.411
 reverse direction: "OL"

2.3 Pin Booster (Trio)

! CAUTION

The PIN Booster contains larger magnetic components such as mains transformer, fan with magnetic housing a.s.o.

There is the risk of physical injury for the service engineer and damage to the components, if it is attracted by the 3T magnetic field.

The following precautions have to be observed:

Minimal distances between the PIN booster and magnet covers (magnetic field is on):

Lateral distance from cover (X-direction) > 0.5-m axial distance from funnel opening (Z-direction) > 1.5 m

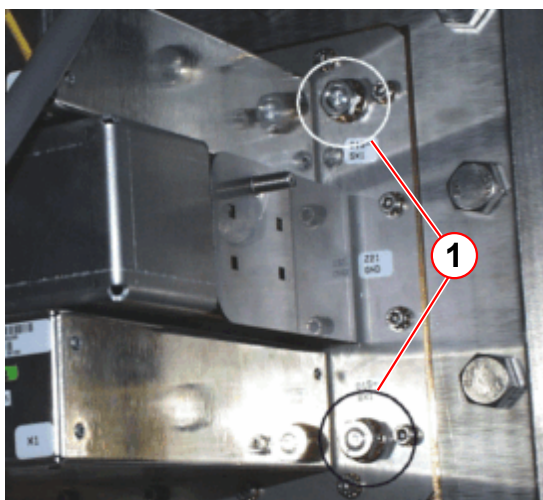
- » Ramp down the magnet in case the PIN booster must be handled close to the magnet in a distance less than 1.5 m.

Fig. 40: Warning label at the PIN Booster



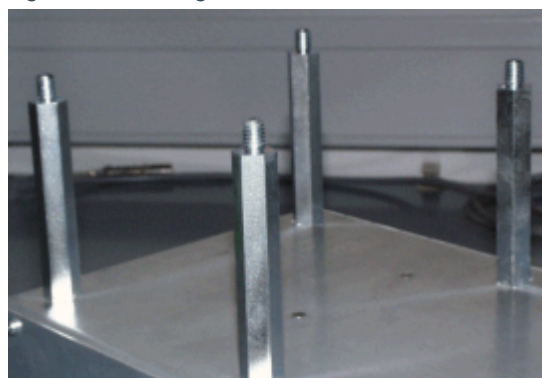
- Switch-off the MR system.
- Remove all cable connections at the PIN booster.
- Remove the 4 mounting nuts at the RF-filter plate in the technical room (refer to (→ Fig. 41 Page 42))

Fig. 41: Two of four mounting nuts (view from technical room)



(1) Mounting nuts

Fig. 42: Mounting studs of PIN booster



- Remove the PIN booster from the RF-filter plate in the magnet room, observing the above described precautions.
- Slide the threads of the mounting studs of the new PIN booster through the holes in the RF-filter plate (refer to (→ Fig. 42 Page 42)). The PIN booster will be self-suspended until you fix the nuts in the technical room.
- Connect the cables and switch on the MR system.

2.4 RFPA LPPA 13080 W (Trio)



CAUTION

Heavy load!

The weight of the RFPA is 55 kg + trolley + shipping crate = 124 kg.

If the RFPA is dropped from the installation height when pulling it out or mounting it in the slide rails, there is potential risk of injury due to its weight. Dropping the RFPA will almost certainly cause damage to the unit

» Observe the (→ Procedure / Page 44).

2.4.1 Special Tools

One set of male and female quick connectors (material number 5762443)

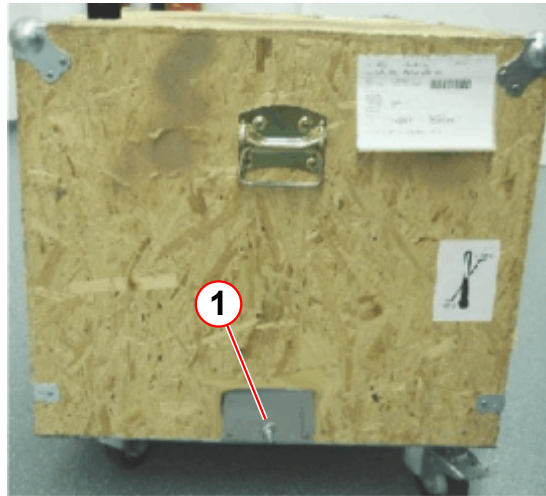
Ratchet with hex/Allen sockets (automobile size)

2.4.2 Procedure

- Inspect the tilt watch (→ Fig. 44 Page 45) on the transport box and follow the instructions.
- Remove the 2 transport box bolts (→ Fig. 43 Page 45) in the front and the rear and take off the wooden lid.
- On the component sticker, verify the revision level of the new RFPA. It must be G00 or higher (→ Fig. 45 Page 45). If not, contact your UPTIME Service Center before continuing.
- Remove the bottom transport box board (4 bolts) from the trolley (→ Fig. 46 Page 45).

- Remove both brackets and set them aside (→ Fig. 47 Page 45).

Fig. 43: RFPA crate



(1) bolt

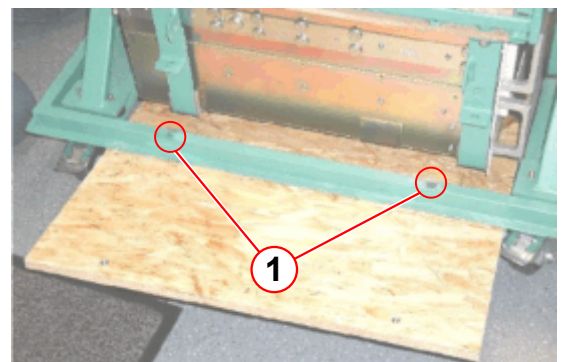
Fig. 44: Tilt watch



Fig. 45: Revision level label

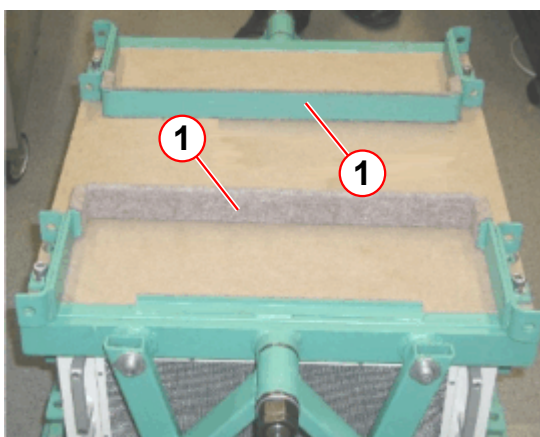


Fig. 46: RFPA bottom particle board



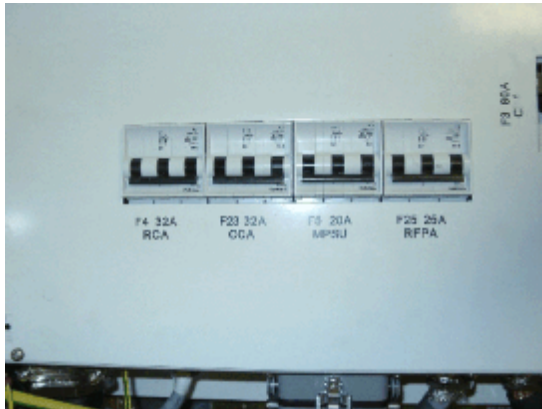
(1) bolts

Fig. 47: Brackets



(1) Brackets

Fig. 48: On-Off switch F23

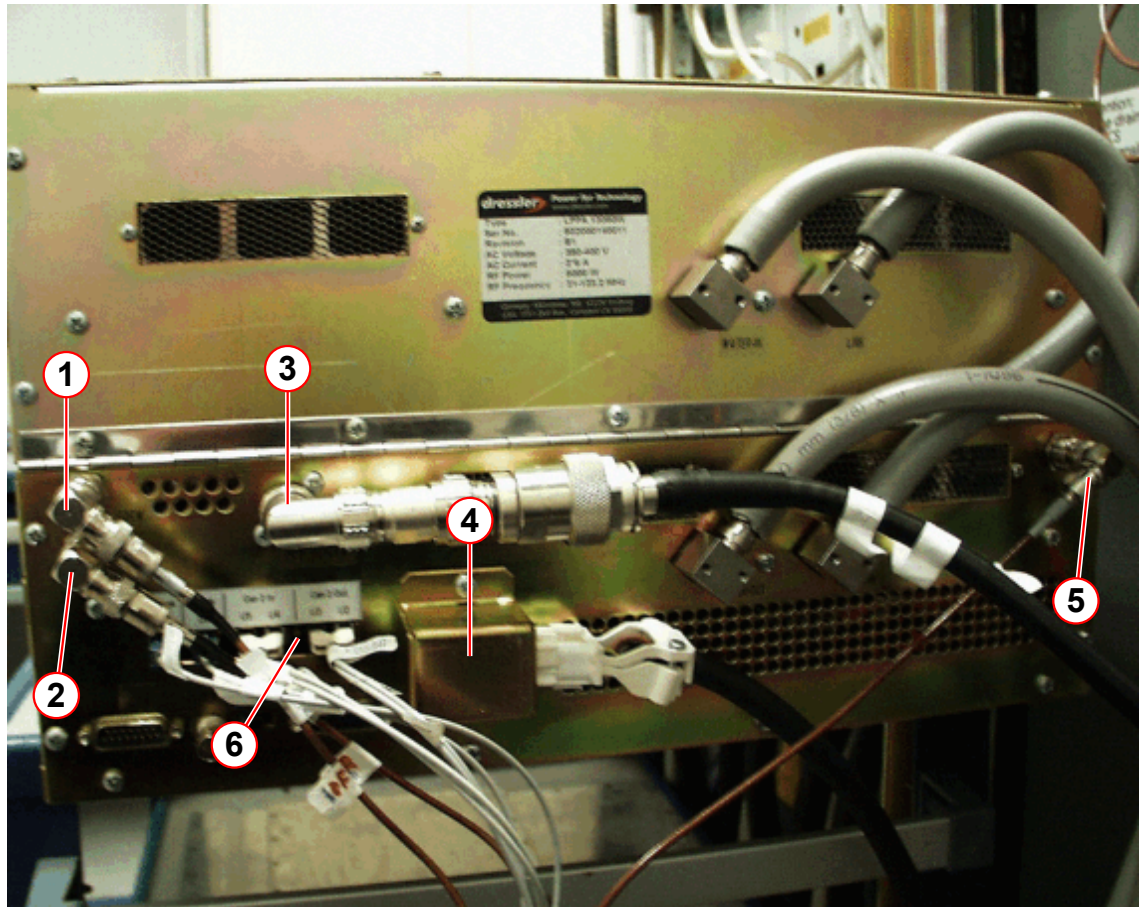


- On top of the CCA switch-off the MR system at the mains box with F23 mains circuit breaker (→ Fig. 48 Page 46).

- Push the trolley with the new Dressier to the front of the CCA (→ Fig. 50 Page 47) .
- If access possible from the rear, disconnect all leads in the back of the RFPA and the gray water hoses (→ Fig. 49 Page 47).
- If access is not possible open the swing frame (→ Fig. 52 Page 48), remove the 4 locking screws for the RFPA and pull it out towards the front (→ Fig. 50 Page 47) until it is latched (→ Fig. 51 Page 47).
- Now disconnect all leads and the gray water hoses:
- Disconnect the following line connectors from the rear side of the RFPA and detach them from their fastening points at the RFPA (→ Fig. 49 Page 47):
 - SSB1 (RF in)
 - PR
 - PF
 - RF out
 - Mains voltage
 - U4/5: FOC CAN2 in (to PTAB)
 - U2/3: FOC CAN2 out (to MMC)
 - U1: FOC RFPA_UNBLANK
 - U6: FOC RFPA_ENABLE

- Disconnect the quick coupling of the cooling water hose in the CCA.

Fig. 49: Connectors at rear side of RFPA



- (1) PF
- (2) PR
- (3) RF out
- (4) Mains power
- (5) RF in
- (6) Unblank-Enable-CAN2in - CAN2out

Fig. 50: Pulling out the RFPA



Fig. 51: Guide rails latch



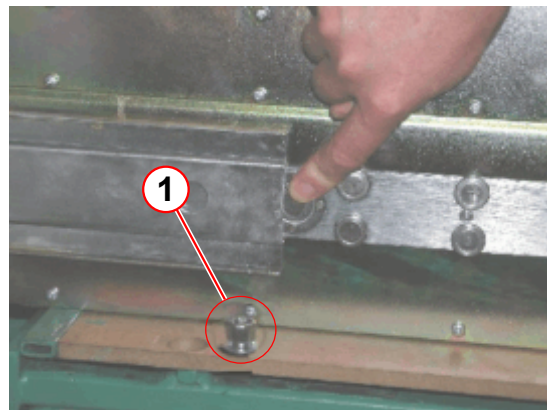


The new RFPA's do not require the water filter anymore. If there is a water filter installed on the old RFPA, discard it locally.

Fig. 52: Swing frame



Fig. 53: Latch and one of four vertical adjustment bolts



(1) bolt

- If the top of the trolley and the bottom of the RFPA are not equal, lower or raise the trolley vertically using the 4 alignment bolts (→ Fig. 53 Page 48) and (→ Fig. 66 Page 52)).
If the alignment bolts are too short, use the longer bolts from the bottom particle board (→ Fig. 46 Page 45).
- Press the latch on both guide rails and pull out the RFPA (→ Fig. 51 Page 47) and (→ Fig. 55 Page 48)) until it sits completely on the trolley (→ Fig. 55 Page 48).

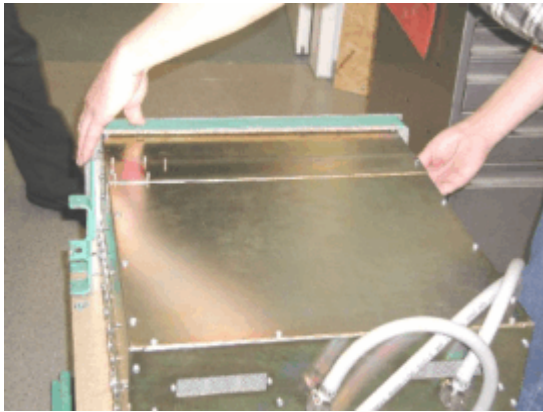
Fig. 54: Guide rails latch



Fig. 55: RFPA resting on trolley



Fig. 56: Bracket



- Secure the old RFPA to the trolley using the 2 brackets and the holes. ((→ Fig. 55 Page 48), (→ Fig. 56 Page 49) and (→ Fig. 47 Page 45))

The brackets must be extremely tightened with an automobile size ratchet. It is imperative that the RFPA cannot move back and forth by its own weight while in transit. It would eventually wreck the RFPA unit if not secured properly.

Fig. 57: End stopper



Assure that the RFPA front is pushed against the end stopper (see arrows (→ Fig. 57 Page 49))

- Drain the water from the RFPA by connecting 2 mating quick connectors (p/n 5762443) to the RFPA water hoses ((→ Fig. 58 Page 49) and (→ Fig. 59 Page 49)).

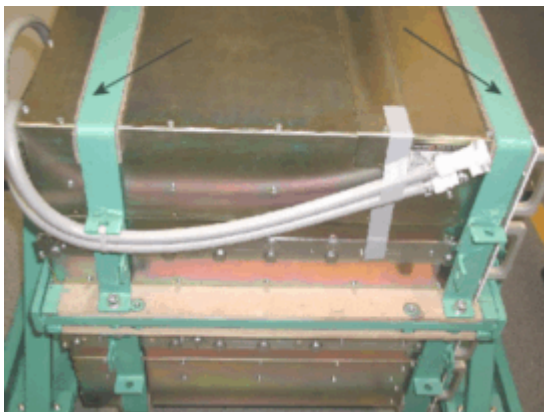
Fig. 58: Draining the RFPA



Fig. 59: Draining the RFPA

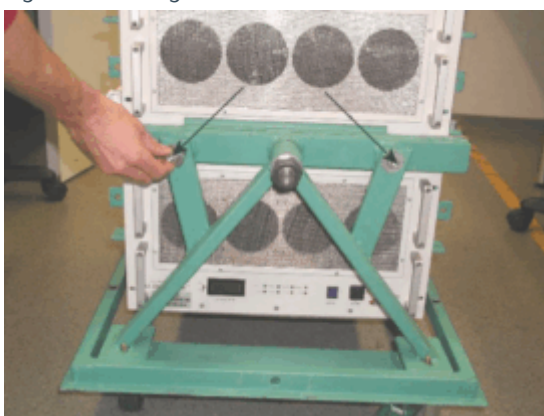


Fig. 60: Old RFPA secured



- Fasten the gray water hoses to the RFPA as shown in (→ Fig. 60 Page 50) using cable ties and/or electrical tape.

Fig. 61: Locking bolts



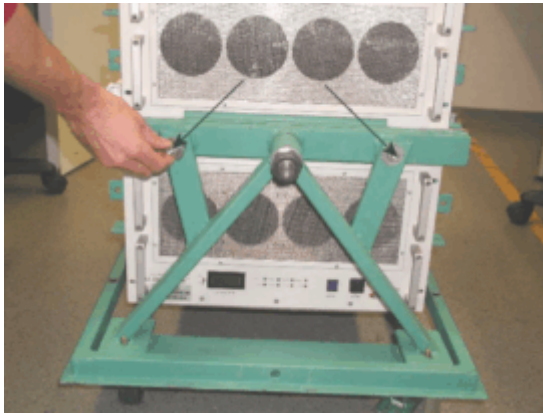
- Remove both locking bolts (→ Fig. 61 Page 50).

Fig. 62: Old (top) and new (bottom) RFPA



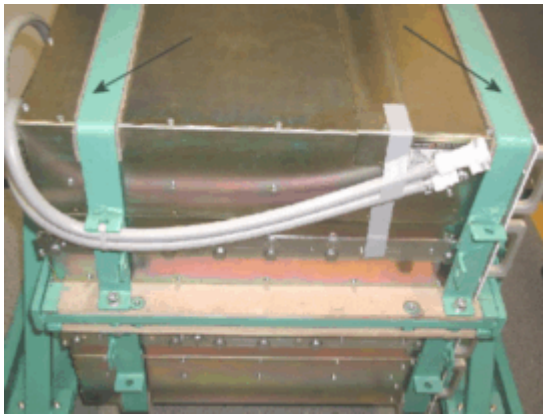
- Rotate the units until the new RFPA is on top.

Fig. 63: New RFPA on top



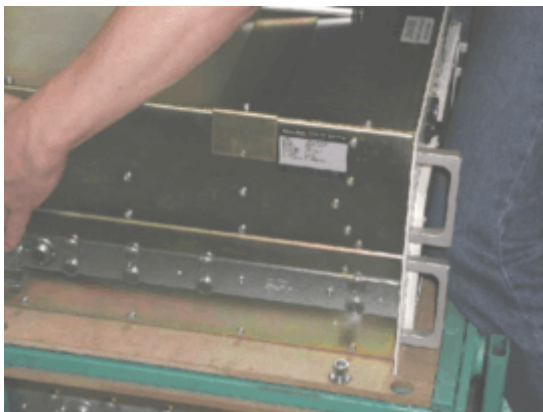
- Insert and **firmly** (extremely important) tighten both locking bolts.

Fig. 64: Brackets



- On the new RFPA, release the water hoses and remove the 2 fixing brackets (→ Fig. 64 Page 51).

Fig. 65: RFPA



- Push the RFPA towards the CCA.

- If the new RFPA and the CCA guide rails are not aligned, turn the 4 vertical alignment bolts clockwise or counterclockwise (→ Fig. 66 Page 52) until both line up ((→ Fig. 67 Page 52) and (→ Fig. 68 Page 52)).

- Use the longer allen bolts from the bottom particle board to align.

Fig. 66: Vertical alignment bolts (two are shown circled)

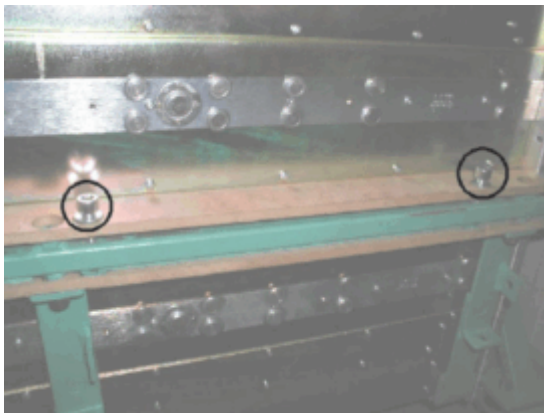


Fig. 67: RFPA and guide rail aligned

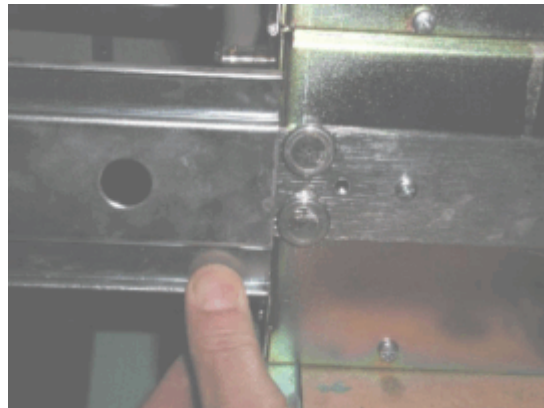
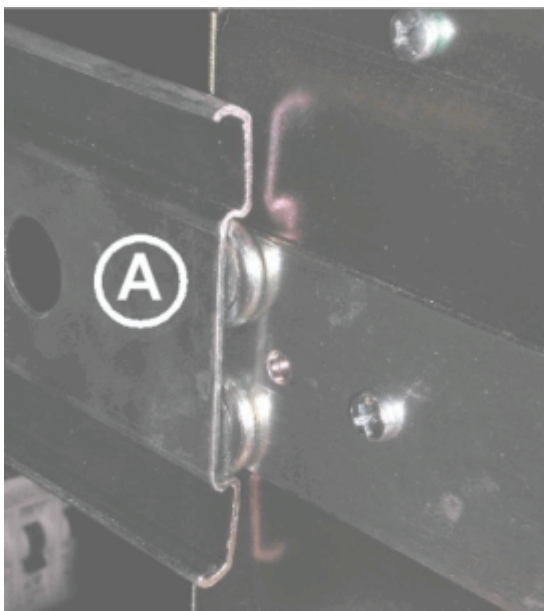


Fig. 68: RFPA aligned

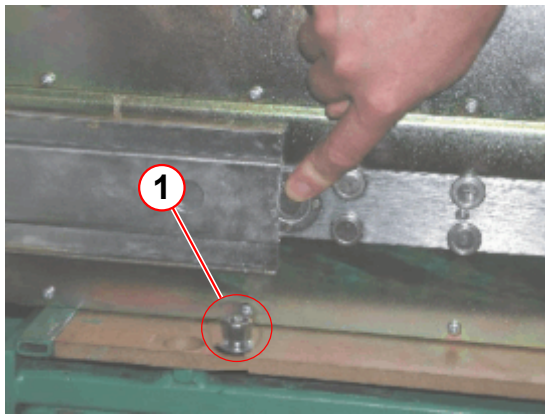


- Push the RFPA in until the latch blocks further movement.
- Press the latch button and continue pushing the RFPA into the guide rail (→ Fig. 53 Page 48).



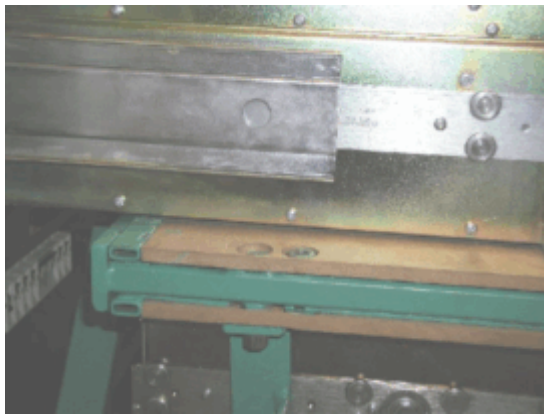
The rail marked as item A in (→ Fig. 68 Page 52) is shipped with the new RFPA and is redundant. It should be returned with the old RFPA to Erlangen.

Fig. 69: Latch and one of four vertical adjustment bolts



(1) bolt

Fig. 70: RFPA pushed in



- Continue until the RFPA latches again (→ Fig. 70 Page 53).

- Reconnect the electrical wires, fiber optics (→ Fig. 49 Page 47) and the water hoses in the back of the RFPA.
- Press the latch button on the guide rails again and push the RFPA carefully into its final position.

Fig. 71: Water hoses and other RFPA leads routed (view from front)



- Route the water hoses appropriately along the CCA right hand side cover and fasten them with cable ties (→ Fig. 71 Page 53).

- Fix the RFPA to the CCA using the 4 fixing screws.

- Remove the trolley with the old RFPA from the equipment room.
- Attach the bottom particle board to the bottom of the trolley (→ Fig. 46 Page 45).
- Attach the 2 spare brackets to the trolley top using the proper threaded holes (→ Fig. 47 Page 45).
- Include the extra guide rails that came with the new RFPA.
- Make sure the RFPA is properly crated in the original shipping box.
- Return the RFPA in the shipping crate.

Fig. 72: RCA top



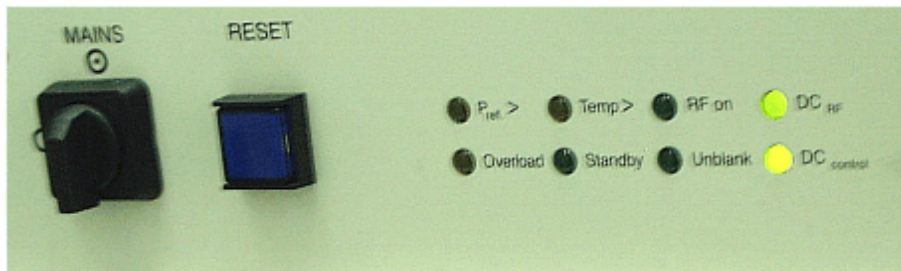
- Check and, if necessary, refill the secondary water circuit with purified water (→ Fig. 72 Page 54).

2.4.2.1 Calibrating the RF Circuit

- Turn-on the CCA circuit breaker F23 and the power switch on the front of the RFPA.
- Switch on the MR system.
 - » After switching on the system and downloading NUMARIS (message "Measurement system ready"), the LEDs on the front side of the RFPA display the following status:
 - "Pref >" must be OFF (if ON after measurement: Power reflected exceeds tolerance limit).
 - "Overload" must be OFF.
 - "Temp >" must be OFF (if ON, check water cooling).
 - "Standby" is OFF (ON only during downloading).
 - "RF on" is OFF; during pulsing with RF, the LED will flicker.
 - Unblank is OFF; during pulsing with RF, the LED will flicker.
 - "DC RF" is ON

- "DC control" is ON

Fig. 73: Mains switch and LEDs on front side of RFPA



- Perform TuneUp / RF Characteristic LC.
- Perform Coil QA / Coil Check with CP Head Coil, RF Verify and Calculation Artefacts.

2.4.3 Final Work Steps

- Increase the CCA revision level to 11.
- Clean up your tools and dispose of all trash.
- Close all cabinets.

2.4.4 Return of Parts

Return the old RFPA using the shipping trolley and transport box (→ Fig. 43 Page 45), and return it to the Goods Receiving department.

2.5 RFPA35kW (Astex /MKS, Trio)

2.5.1 General Safety Information



CAUTION

Heavy load!

Weight of amplifier is approx. 450 kg!

Take care when moving the amplifier in and out of the shipping crate to its final position!

- » Hazardous conditions. Ensure that the amplifier does not tilt over. 2 persons are recommended.
- » Observe the attached "Unpacking and packing instructions".



WARNING

High voltage!

Hazard of electric shock or burn. Even when the MR system is switched off, mains voltage is present at the power input of the amplifier.

- » Before working on the amplifier, switch-off the circuit breakers CB2 and then CB1 at the amplifier and at least circuit breaker main circuit breaker/25A in the mains box (on top of CCA) and ensure against accidental switch-on.
- » Observe the warning information on the amplifier.
- » After the amplifier has been turned off, allow at least 10 minutes for the high voltage filter capacitors to discharge fully before removing any of the cover plates.

2.5.2 RFPA35kW Replacement

Material number 73 86 951

2.5.2.1 Tools and time required

Time

60 min

Tools

Normal tool set.

2.5.2.2 Preliminary

1. Place the amplifier in OFF mode. Amplifier fans will remain "ON" for approximately 1 minute to allow amplifier to cool down.
2. After the fans stop, switch-off the line circuit breakers (CB2, CB1 at amplifier PS deck and main circuit breaker in the mains box on top of CCA - in that order).
3. Allow at least 10 minutes for all high voltage power supply units to discharge.

2.5.2.3 Replacement Procedure

Disassembly the old RFPA

- Disconnect the mains power cable.
- Disconnect the RF power cable, the cable RF_IN, PR and PF and all FOCs.

Assembling the new RFPA

- Observe the "Unpacking and packing instructions".
- Connect the power , RF, and FOC cables.
- Check the mains accommodation according to (Mains accommodation of new RFPA).
- Route and secure the cables according to their position at the old RFPA.

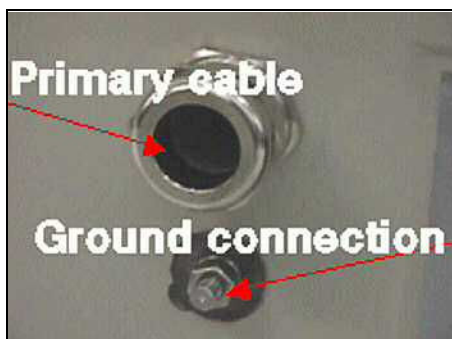
2.5.2.4 Mains Accommodation of new RFPA

The amplifier comes with a variable tap transformer that can be set for the following VAC input settings: 380 / 400 / 420 / 440 / 460 / 480 $\pm 10\%$, 50 / 60 Hz. The tap selected must agree with the site voltage. When shipped from the factory, the system is set up for 480 VAC operation.

There are 2 mains power circuit breakers (CB1/CB2; 20A) located on the front of the Power Supply (lower) deck.

The main AC power connection is located in back of the Power Supply (lower) deck. Refer to (→ Fig. 75 Page 58) for connection details.

Fig. 74: Mains Power Connection

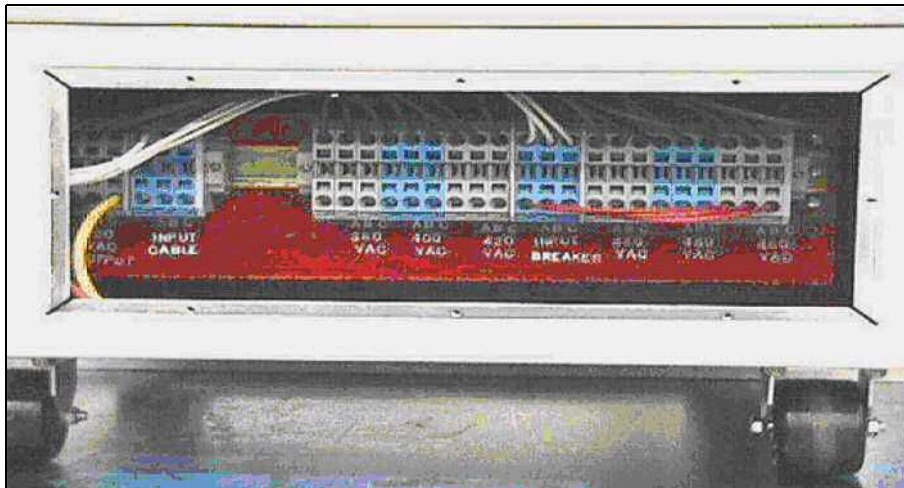


An EMI filter is switched in series with the mains circuitry to the remainder of the amplifier.

You can re-tap the transformer for the VAC input setting required for your site as shown in (→ Fig. 75 Page 58). The wire blocks are color-coded and grouped in sets of 3 (1 block per wire, 3-phase power).

- Determine the site voltage.
- Disconnect the three wires going to the 480 VAC taps (A, B, & C) shown as "L1-L2-L3" and leave the other end of the wires connected to the Input circuit breaker taps. To remove each wire insert a flat-blade screwdriver into the opening directly above the wire to be removed. The blade will rest against a wire release "key". Push the blade against the key until the wire is released. The wire can then be removed.

Fig. 75: AC Mains Input and Transformer Re-Tap Connections



- Reconnect the 3 wires removed from the 480 VAC taps (A, B, & C) to the correct taps for your site voltage (A, B, & C). Note each "block" is color-coded and arranged in sets of 3. To insert each wire in the new tap, insert a flat blade screwdriver into the opening directly above it where you want to install the wire. The blade will rest against a wire release "key". Push the blade against the key until the wire can be inserted. Once the wire is inserted, remove the screwdriver blade; the key will then lock the wire in place.
- Connect the VAC input cable to the transformer marked input cable taps. A 0.51 to 0.71 inch (13 to 18 mm) diameter power cable is required for safety reasons.

2.5.2.5 Concluding Steps

- Prepare the old RFPA for shipment in the transport box. Observe the "Unpacking and packing instructions".
- Switch ON the main circuit breaker in the mains box on top of CCA, CB1 and CB2 at the amplifier PS deck - in that order.
- Switch on/ reboot the MR system/scanner.

2.5.2.6 Adjustments

- Perform TuneUp/ RF Characteristic
Time 5 min

2.5.2.7 Final Checks

- Perform QA / RF Verify
 - Perform QA / Calculation Artefacts
 - Perform QA / Stability_Long Term Check (if available)
- Time 30 min

2.5.3 PA/ IPA Tube Replacement

Part no:	IPA Tube:	7388809
	PA Tube:	7388791

2.5.3.1 Safety Information



WARNING

Lethal voltages are present inside the power supply and RF decks!

Hazard of electric shock or burn. Even when the MR system is switched off, mains voltage is present at the power input of the amplifier.

- » Before removing any equipment cover, turn off the equipment, unplug or disable the AC mains supply and wait at least ten minutes to allow the power supply units to discharge fully.



WARNING

The PA/IPA tube may implode.

Risk of serious physical injury!

- » Use protective clothing such as: - Ear protection (sound reduction factor 26 dB(A) - leather gloves - Neck and wrist protection - face protection shield

2.5.3.2 Tools and time required

Time

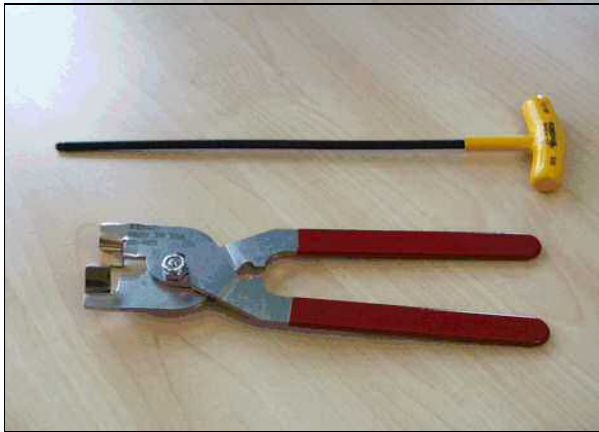
60 min

Tools

- Cross-type (Philips) screwdriver, #2 size
- Blade-type screwdriver, 4-mm width
- MKS-supplied tube-pulling pliers - use only for IPA tube removal, not installation! (XWX-X113)

- MKS supplied hex wrench - use for PA tube removal and installation (XWX-X117).
- Torque hex wrench, 5/32-inch (4mm)
- Torque hex wrench, 7/16-inch (11 mm)
- Discharge rod (supplied, connected to the amplifier).

Fig. 76: Tube-pulling pliers and hex wrench (supplied with the amplifier)



2.5.3.3 Preliminary

1. Place the amplifier in OFF mode. Amplifier fans will remain "ON" for approximately 1 minute to allow amplifier to cool down.
2. After the fans stop, switch-off the line circuit breakers (CB2, CB1 at amplifier PS deck and main circuit breaker in the mains box on top of CCA - in that order).
3. Allow at least 10 minutes for all high voltage power supply units to discharge.

There are many system components that are static sensitive. Take appropriate precautions to avoid damage to these sensitive components.



Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

2.5.3.4 Replacement Procedure

- Remove the RF deck top cover and tube compartment cover.
 - » After removing the tube compartment cover, both the IPA and PA tubes are visible.

Fig. 77: Tube Compartment Cover Removal



⚠ WARNING

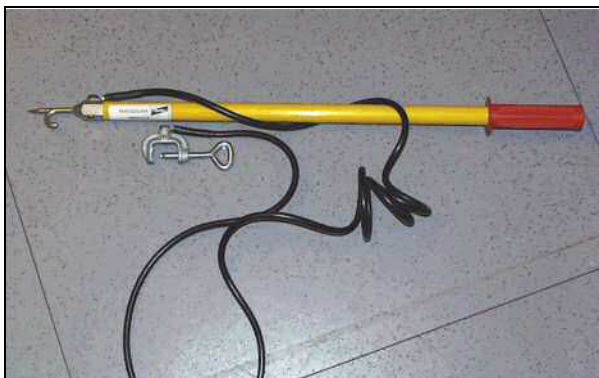
High voltage may remain at the tubes and other components in the compartment. If ignored, death or serious physical injury can occur.

- » Do not touch anything in the compartment until the high voltage is discharged using the discharge rod (→ Fig. 78 Page 61) as follows:



Note the special hardware that "captivates" the tubes. Do not remove any of the hardware or circuit components until you are directed to do so in the following steps. Each tube may be removed independently. It is not necessary to remove both tubes in order to replace just one.

Fig. 78: Discharge rod, cable and clamp for connecting to chassis ground



2.5.3.5 Discharging of Components in the Tube Compartment

- Discharge the tube anodes and its plate choke using the discharge rod which must be connected via cable to ground (amplifier chassis). Refer to (→ Fig. 79 Page 62) and (→ Fig. 80 Page 62).

While working inside the tube compartments, keep the discharge rod hooked to one of the marked points to avoid recharging.

Fig. 79: PA tube compartment - Locations to be discharged (Beta-type)

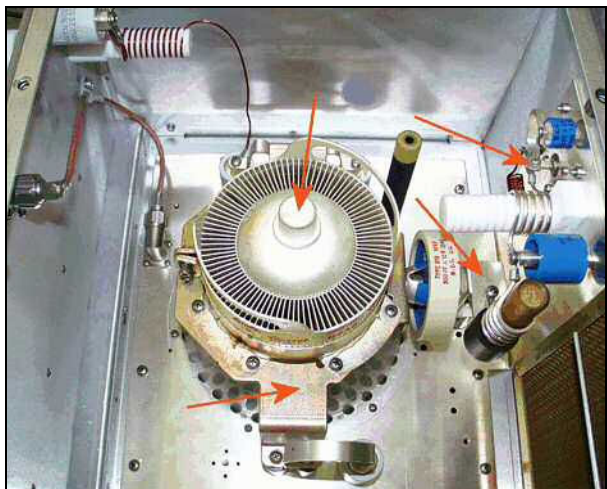
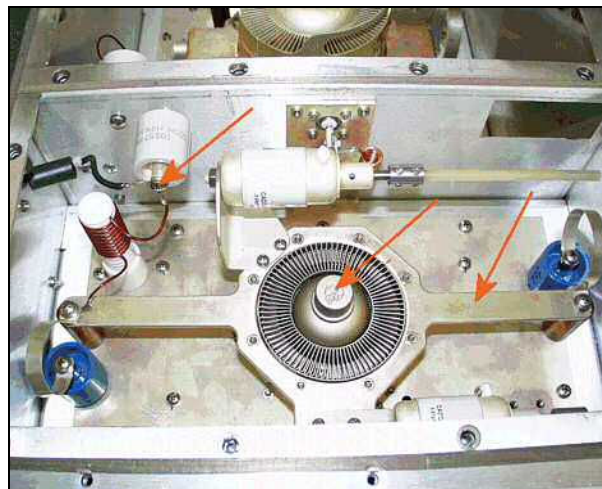


Fig. 80: IPA tube compartment - Locations to be discharged (Beta-type)



2.5.3.6 PA Tube Clamps

Connection to the PA tube anode is made via a ring bracket that surrounds the anode. The bracket is secured to the tank circuit with 8 bolts (aligned in 4 "sets" of 2) in the shape of a "square diamond".

The original version consists of 1 ring bracket that is clamped around the tube with 2 bolts. The new version consists of 2 mechanical pieces. A circular bracket makes the connection between the tube anode and the tank circuit. A band clamp secures the bracket to the tube anode (→ Fig. 81 Page 62).

Fig. 81: Tube clamps, left original, right new style



2.5.3.7 PA Tube Removal

Connection to the PA tube anode is made via a ring bracket that surrounds the anode. The bracket is secured to the tank circuit with 8 bolts (aligned in 4 "sets" of 2) in the shape of a "square diamond".

2.5.3.8 Tube Clamp Removal (Original Style)

- Remove the 8 bolts from the ring bracket.

- With a short 5/32 inch (4 mm) hex wrench, loosen the 2 horizontally-mounted hex "clamping bolts" that hold the ring bracket around the tube. The bolts are shown in close-up in (→ Fig. 82 Page 63) and (→ Fig. 83 Page 63).

Fig. 82: PA Tube Close-Up

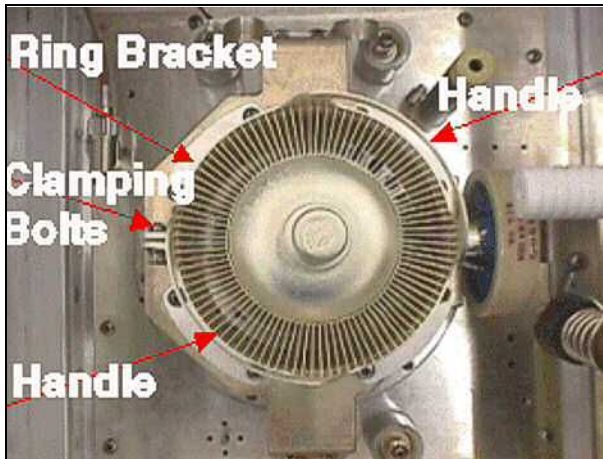
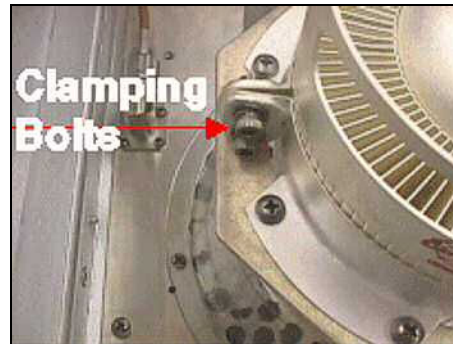


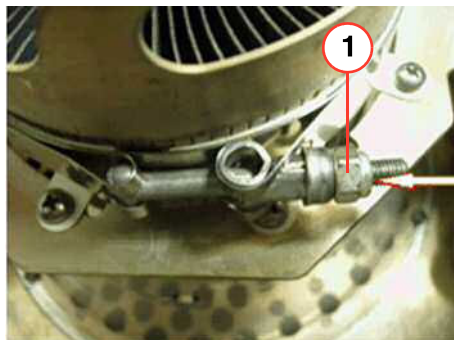
Fig. 83: Ring Bracket Clamping Bolts, Close-Up



- Lift and remove the ring bracket from the tube anode.

2.5.3.9 Tube Clamp Removal (New Style)

Fig. 84: Band clamp securing nut



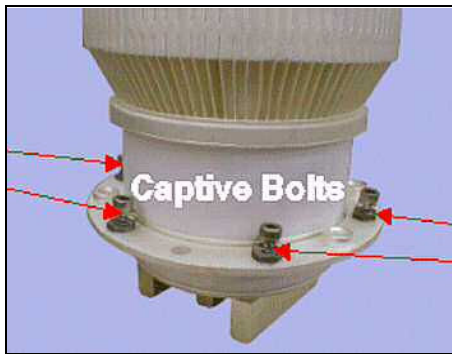
(1) Securing nut

- Using a 7/16 inch (11 mm) wrench, loosen the nut securing the band clamp to the tube until the clamp can be easily removed (→ Fig. 84 Page 63).
- Remove the 8 bolts securing the bracket to the tank circuit.
- Lift the bracket out of the tube compartment.



After the ring bracket has been removed, 6 access holes located around the perimeter of the tube will be visible in the tank circuit inductor. The tube-securing captive bolts will be loosened as follows: The tube-securing captive bolts are shown in (→ Fig. 85 Page 64).

Fig. 85: PA Tube Base Showing Captive Securing Bolts



The grid ring around the circumference of the tube base has 5 spring-loaded hex-head bolts. These allow the tube to be secured to the socket from the top of the compartment. A specially-modified hex wrench is supplied with the amplifier, part XWX-X117, and is shown in (→ Fig. 76 Page 60).

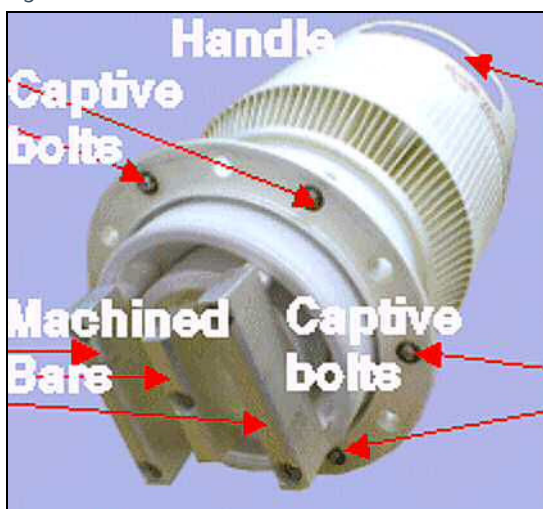


The wrench allows the bolts to be loosened or hand-tightened as needed.

- Use the supplied hex wrench (XWX-X117) to loosen all 5 of the hex-head bolts. Note the 5 bolt access holes are located in the tank circuit inductor that surrounds the perimeter of the tube. Insert the wrench in any hole then extend it downward and into the hex-head bolt. Loosen the bolt by turning it to the left (counter-clockwise) until the bolt head "pops" up. Loosen the remaining bolts.
- After all 6 bolts are loosened and are completely free of the chassis carefully lift the tube - with its built-in "handles" -- STRAIGHT UP and out of its socket. **DO NOT ROTATE THE TUBE!**

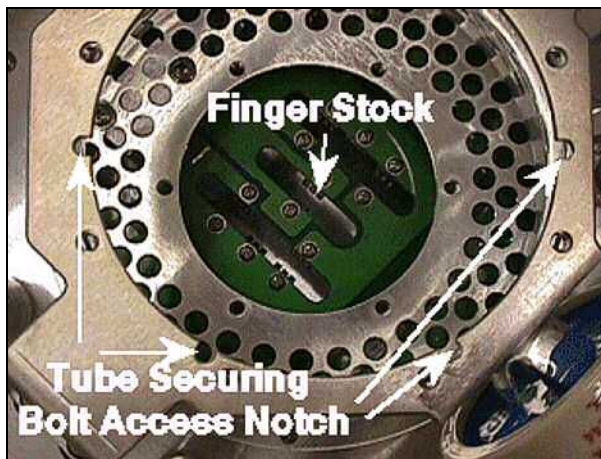
2.5.3.10 PA Tube Installation

Fig. 86: PA Tube Base



The base of the PA tube has specially machined bars attached to it. The bars allow easy tube installation from the top of the compartment. The companion socket has been modified with "finger stock" which, when the tube is inserted, will contact the tube filament and cathode bars. The PA tube base is shown in (→ Fig. 86 Page 64).

Fig. 87: PA Tube Socket



The PA tube socket is shown in .
(→ Fig. 87 Page 65)

■ **INSPECT THE SOCKET - ENSURE NONE OF THE FINGER STOCK IS DAMAGED!**

To install the PA tube follow these steps:

- Don't touch the tube without wearing leather gloves.
- Lift the tube with its handles and orient the tube base "bars" with the socket finger stock.
- Carefully lower the tube into its socket while ensuring that the base bars will mate with the finger stock.
- Press the tube downward into the socket.
- Use the supplied hex wrench (XWX-X117) to start and thread the 6 captive hex-head bolts attached to the tube base ring into their respective chassis retaining nuts. **Do not allow the screws to cross-thread.**
- Use the supplied hex wrench (XWX-X117) to tighten the hex-head bolts on the tube base ring.

Tube Clamp Installation (Original Style)

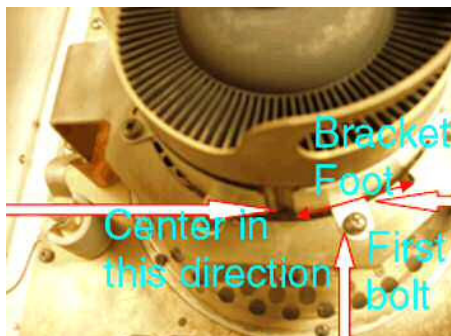
- Refer to (→ Fig. 79 Page 62). Replace the tube anode ring bracket with the opening of the bracket facing the IPA tube compartment; start all 8 bolts and hand tighten them (do not torque).
- Using a 5/32 (4 mm) torque hex wrench, tighten the 2 clamp bolts to secure the bracket to 8in-lbs (1.0+/-0.1Nm). Repeat the torque 4 times, alternating between the bolts.
- Tighten the 8 bracket bolts surrounding the tube down to 23 in-lbs (2.6+/-0.1 Nm).

Tube Clamp Installation (New Style)

- Replace the tube anode ring bracket with the opening of the bracket facing the IPA tube compartment; start all 8 bolts and hand tighten them (do not torque).
- At the point of the first bolt to the right of the bracket opening (→ Fig. 88 Page 66), push the bracket against the tube anode and center the foot (side to side) under the

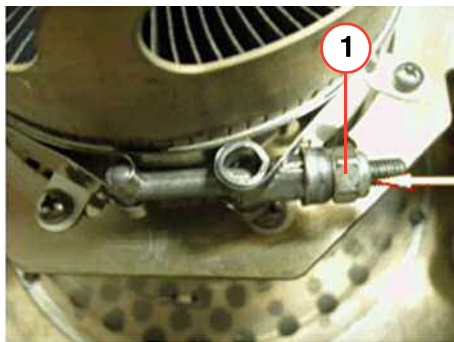
bolt. While pushing the bracket against the tube, torque this bolt to 23 in-lbs (2.6+/-0.1 Nm only).

Fig. 88: Tube clamp installation



- Slide the band clamp over the tube bracket oriented as shown in (→ Fig. 89 Page 66) and tighten the nut down until the band clamp will stay in place although it can still be moved by applying manual pressure.
- Rotate the clamp to the approximate position shown in (→ Fig. 89 Page 66).

Fig. 89: Band clamp securing nut



(1) Securing nut

- Align the top edge of the band clamp until it is flush with the top edge of the tube bracket all the way around the tube.
- While keeping the clamp and bracket flush together, torque the nut down to 39 in-lbs (4.5+/-0.1 Nm).
- Torque the remaining 7 bracket bolts down to 23 in-lbs (2.6+/-0.1 Nm).

2.5.3.11 Final Steps

When tube installation is complete, replace the tube compartment and RF deck top covers, reconnect the system cabling as appropriate, and test. The tube hours counter should be reset after tube replacement. The reset procedure is described in section (→ Tube Hours Counter Reset / Page 69). The test procedure is described in section (→ Adjustments / Page 68) and (→ Final Checks / Page 69).

2.5.4 IPA Removal and Installation

The IPA tube portion of the RF deck is shown in (→ Fig. 80 Page 62).

High voltage connection to the IPA tube is made via a ring bracket that surrounds and contacts the anode. The bracket is secured to the tank circuit and is not to be removed! The tube is removed from its socket with a specially designed set of pliers (XWX-X113) shown in (→ Fig. 76 Page 60).

The pliers are supplied with the amplifier. They are designed to grip the tube anode cap only.

Follow this single step:

- Use the pliers to grip the tube anode cap, then pull the tube **STRAIGHT UP** and out of its socket. The socket fit will be tight. **DO NOT ROTATE THE TUBE DURING REMOVAL!**

The IPA tube removal technique is illustrated in (→ Fig. 90 Page 67).

Fig. 90: Tube Removal Technique

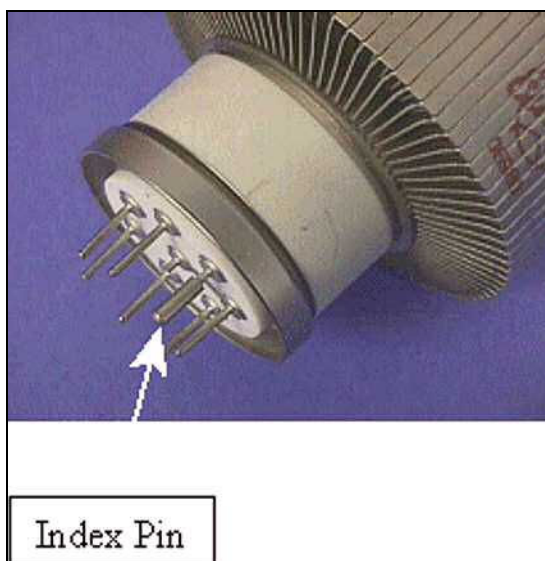


To install the IPA tube follow these steps:

- Don't touch the tube without wearing leather gloves.
- Verify that all base pins are straight; straighten them (with long-nose pliers, not supplied) if necessary.

- Locate the IPA tube "index" pin. It is larger than the others. The index pin is shown in (→ Fig. 91 Page 68).

Fig. 91: IPA Tube Base Pins - Index Pin Highlighted



- Orient the tube index pin with its socket mate. Lower the tube straight down into the chimney.
- Carefully rotate the tube side-to-side, as needed, to align the index pin with its socket mate. Once properly oriented, the tube will partially "drop" into the socket.
- Apply even, vertical, downward pressure on the tube anode to seat the tube in the socket. Don't rotate and press down on the tube at the same time.

2.5.4.1 Final Steps

When tube installation is complete, replace the RF deck top cover, reconnect the system cabling as appropriate and test. The tube hours counter should be reset after tube replacement. The reset procedure is described in section (→ Tube Hours Counter Reset / Page 69). The test procedure is described in section (→ Adjustments / Page 68) and (→ Final Checks / Page 69).

2.5.4.2 Concluding Steps

Switch ON the main circuit breaker in the mains box on top of CCA, CB1 and CB2 at amplifier PS deck - in that order.

Switch on the MR system.

2.5.4.3 Adjustments

- Perform TuneUp/ RF Characteristic.
Time 5 min



The fault 67, "IPA Tube Arc" could happen 1 or 2 times after RF tube replacement. This behavior is normal according to the manufacturer.

Reboot scanner if the error occurs.

2.5.4.4 Final Checks

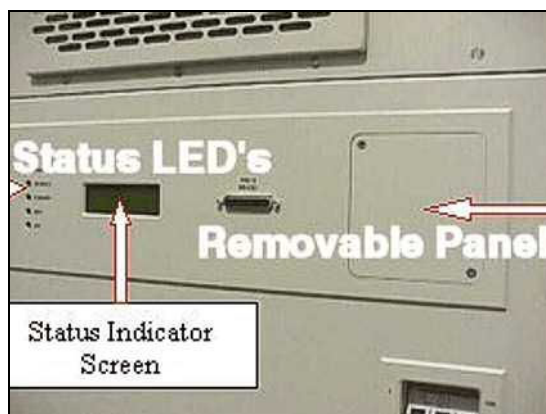
- Perform Quality Assurance/ RF Verify.
- Perform Quality Assurance/ Calc Artefacts.
- Perform Quality Assurance/ Stability_Long Term Check (if available).

Time 30 min

2.5.4.5 Tube Hours Counter Reset

The tube hours counter reset is done manually via the amplifier system controller keypad. The keypad is located on the system controller circuit board. It is accessed via the removable cover located on the status panel on the front side of the amplifier. The panel is shown in (→ Fig. 92 Page 69).

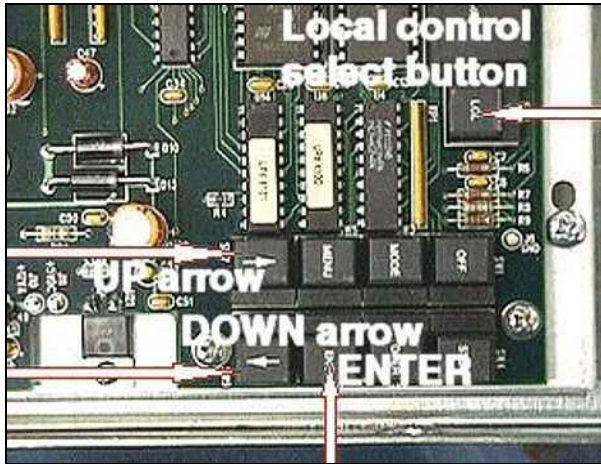
Fig. 92: Status indicator screen at front panel



Follow these steps:

- The panel is secured with 2 cross-blade screws at the upper left and bottom right corners of the panel. Remove these 2 screws. Once the panel is removed, the keypad behind it will be visible. The keypad is shown in (→ Fig. 93 Page 70).

Fig. 93: Controller keypad



- Turn ON the line circuit breakers CB-1 and CB-2. (Note: Mains power must be applied to the amplifier in order to reset the tube hours counter. However, the amplifier itself can be Off, in Stand-by, or in the Operate mode at the discretion of the service technician.)
- Depress the LCL ("local" control) button on the System Control board.
 - » A green LED within the button should illuminate. (host computer commands will now be ignored.)
 - » The status indicator window should display menu 1. A cursor will appear to the left of one of the four line entries.

Menu 1

AMP HOURS (Total accumulated amplifier hours)

PA TUBE (PA tube accumulated hours)

IPA TUBE (IPA tube accumulated hours)

SITE HRS (site accumulated hours)

- Depress the arrow buttons, "up" or "down" as shown in (→ Fig. 93 Page 70) to select the counter to be reset.
 - » The cursor will move accordingly as either arrow button is depressed.
- Once the counter has been selected (the cursor will appear to the left of the counter name, as shown in Menu 1), depress the ENTER (ENT) button.
 - » The chosen reset menu, either menu 2, menu 3, or menu 4, will appear on the indicator window.

Menu 2

ZERO PA TUBE HOURS

Menu 3

ZERO IPA TUBE HOURS

Menu 4

ZERO SITE HOURS

- Depress the ENTER button to set the selected counter to zero.
- If another counter is to be reset, select it with the arrow buttons.
- Depress the ENTER button to set the selected counter to zero.
- After all resets are done, depress the LCL ("local" control) button. (This returns control back to the Host.)
- Replace the panel (removed in Step 1) and reinstall the securing hardware.

2.5.5 Fuse Replacement

All amplifier fuses are located on a fuse board in back inside the power supply deck. Access to the fuse board is via a removable panel. The panel has a directly visible fuse location "map" printed on it. The panel is mounted above the mains primary connection point and is shown in (→ Fig. 94 Page 71).

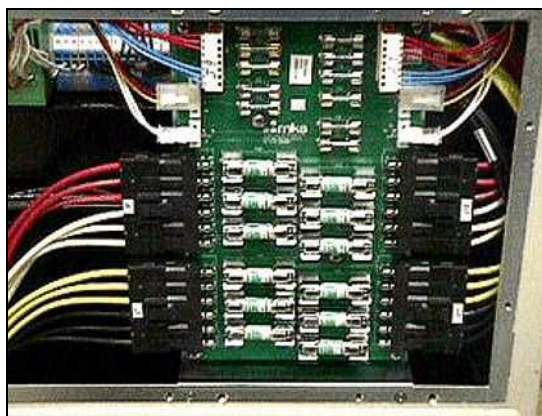
Fig. 94: Fuse location map



Some amplifier fault descriptions list a condition caused by no- or low-supply voltages. These faults may be caused by a defective fuse. Check the appropriate fuse before replacing any other FRU.

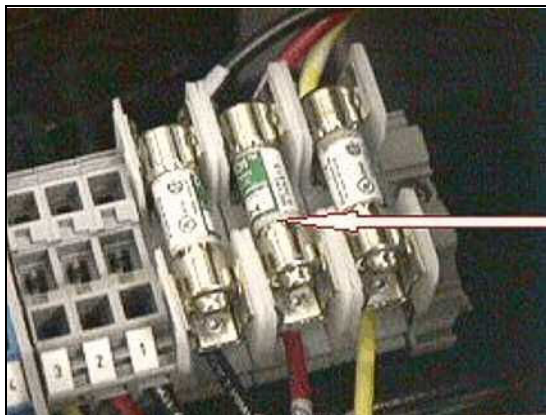
A spare fuse kit is packed with the amplifier. The kit contains replacements for each fuse used within the amplifier. The fuse board is shown in (→ Fig. 95 Page 71)

Fig. 95: Fuse board



A smaller second block is visible above and to the right of the fuse board when the cabinet's rear panels are removed. This block contains only 3 fuses. **These fuses are NOT field replaceable.** The block is shown only for identification. (→ Fig. 95 Page 71) shows the smaller second fuse block.

Fig. 96: These three fuses are NOT field-replaceable



The fuses are identified by designator, function, and current rating.

Fuse Designation	Circuit Function	Function Current Rating
F1, F2	Blower	4 Ampere
F3, F4, F5	Phase reference	250 milliampere
F6, F7, F8	- 15 volt supply	4 Ampere
F9, F10, F11	+ Bias Supply	3.15 ampere
F12, F13, F14	+ 15 volt supply	10 ampere
F15, F16, F17	+ 56 volt supply	15 ampere
F18, F19, F20	PA filament supply	20 ampere
F21, F22, F23	IPA filament supply	20 ampere

2.5.5.1 Deck Separation/Recombination

Part no:	PS deck	73 89 799
	RF deck	73 89 781



Prerequisites for replacing individual PS/RF decks:

Rev.Status 05 is checked off on the ASTEX/ MKS RFPA

Rev.Status 05 is checked off on the PS deck

Rev. Status 05 is checked off on the RF deck.

On the spare part as well on both decks of the RFPA system, the following label has to be attached:

FRU-compatible Caution:

This deck may only be replaced when both decks are labeled FRU compatible.

Tools and Time required

Time

60 min

Tools

- Cross-type screwdriver, #2 size
- Blade-type screwdriver, 1/8 inch (3 mm) width
- Blade-type screwdriver, 1/4 inch (6 mm) width

Preliminary

- Turn OFF the amplifier (CB2, CB1 at amplifier PS deck and main circuit breaker in the mains box on top of CCA -in that order) and wait at least 3 minutes to allow for high voltage discharge.
- While the high voltage is discharging, disconnect all cabling, including the mains supply, from the amplifier.
- The individual decks, the upper RF and the lower power supply, must be separated from each other for individual replacement. A second person is required for separating the decks.

2.5.5.2 Deck Separation

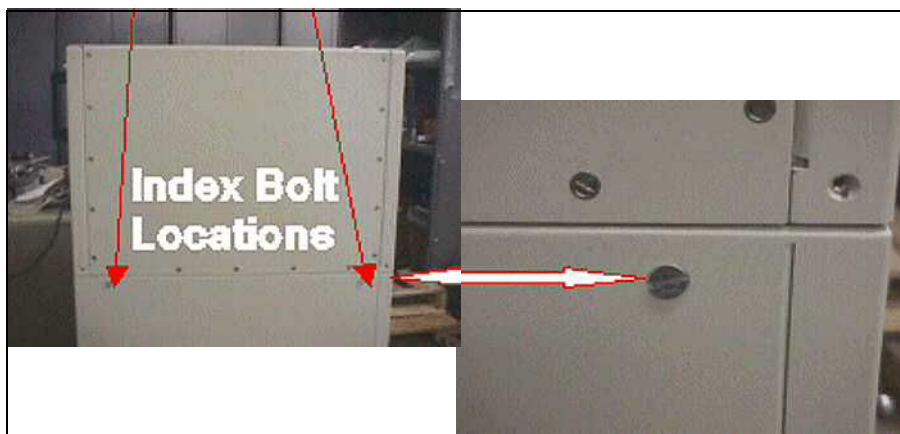
To separate the decks, proceed as follows:



The RF (upper) deck is essentially a rectangular box. Along its bottom edge, at each corner, it is fitted with short length alignment block "legs". These alignment blocks protrude downward. The alignment blocks serve 2 functions. They provide indexing anchor points to secure the upper RF deck to the lower power supply deck, and, when the deck is removed, they serve as legs to allow the deck to be set on a flat floor or table without causing damage to critical wire harness connectors.

- Locate and remove the single indexing bolt from each of the four RF deck bottom-corner alignment blocks. They are shown in (→ Fig. 97 Page 74).

Fig. 97: Close-up (corner index bolt)



- Remove the top cover from the RF deck.
- Remove the left-side panel from the RF deck. Several RF deck wire harness connectors must be unplugged from mating sockets located on the top left corner of the power supply deck. The connectors are shown in (→ Fig. 98 Page 74). The yellow bands (towards outside) mark the correct position of the connectors.

Adjacent to the connectors is a single insulated green-yellow wire attached to a chassis stud. This wire must also be removed. It is shown in detail in (→ Fig. 99 Page 74).



If the yellow bands are missing, mark the position of the connectors to avoid wrong connections.

Fig. 98: Inter-Deck Connectors

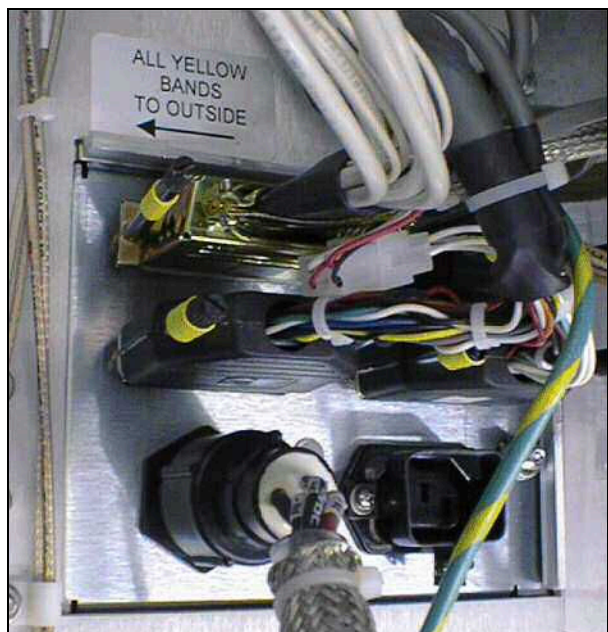
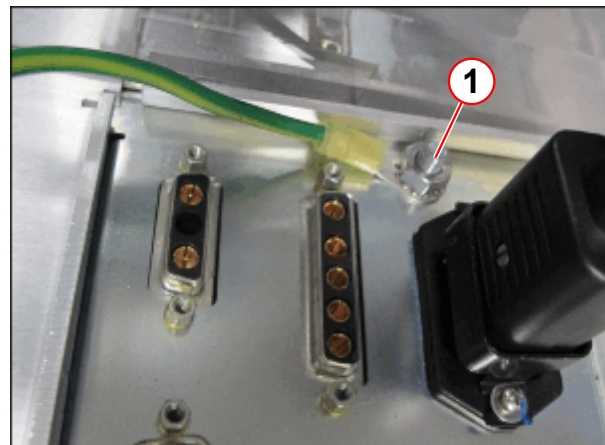


Fig. 99: Ground wire detail



(1) Ground wire stud

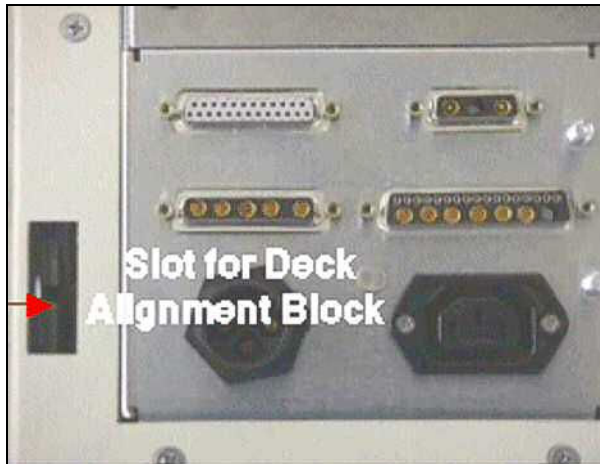


Once the connectors are unplugged, push the cable bundles away from the sharp edges of the RF deck chassis cutout. Fix them temporarily with cable ties to avoid that they can drop back through the chassis cutout and get damaged when the RF deck is set down..

- Lift-off the RF deck from the PS deck in a straight direction. Use the handles which are mounted front- and rear-side. Carefully set the RF deck aside.

2.5.5.3 Deck Recombination

Fig. 100: Power Supply Deck Harness Connectors



A single connector panel is located on top of the power supply deck: RF deck harness plugs are inserted into the panel after the decks are combined. The panel is shown in (→ Fig. 100 Page 75).

Also shown in (→ Fig. 100 Page 75) is an adjacent slot that accommodates one of the four-RF deck alignment blocks. (The blocks are located at each bottom corner of the RF deck.)



The connector receptacle pins are fragile and can get damaged easily. Inspect them carefully before recombining the RF deck to the PS deck. **PAY ATTENTION TO CONNECT THE PLUGS IN THE RIGHT ORIENTATION**, otherwise pins might get damaged and the whole deck must be replaced

- Remove the top- and left-side covers from the RF deck cabinet.
- Lift the RF deck and place it accurately on top of the power supply deck. **The RF deck index bolts blocks must be properly inserted into their respective slots on top of PS deck.**
- Re-install the index bolt locking screws.

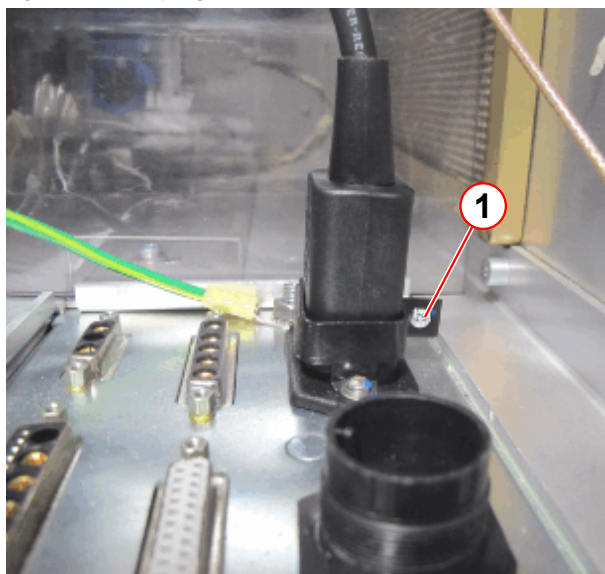
NOTICE

Potential connecting problem of the fan power plug. (RFPA- Fault 41: PA AIR FLOW)

Improper plugging of the fan power connector will lead to a fan error (Fault 41).

- » Open the securing clamp screw before plugging the fan connector.
- » Make sure that the plug is fully engaged before fastening the securing clamp screw.

Fig. 101: Fan plug



(1) Screw of the secured clamp

- Make sure that the securing screw (Pos. 1) is released.
- Check the conditions of the rubber strip glued on the inner side of the securing clamp. This rubber strip is to entirely keep the plug inside the socket even under worst case vibration conditions.
- Push the fan power plug firmly into the socket. You have to overcome the resistance of the rubber strip.
- Tighten the screw (Pos. 1) to avoid loosening of the plug due to vibrations.

- Connect the remaining RF deck plugs to the appropriate receptacles at the PS deck panel.
- Re-install the left- and top cover on the RF deck.

Concluding Steps

- Install all system cabling, including the line voltage to the amplifier.
- Switch ON the main circuit breaker in the mains box on top of CCA, CB1 and CB2 at amplifier PS deck - in that order.
- Switch the MR system "ON" at the Alarm Box.

Adjustments

- Perform TuneUp/ RF Characteristic.
Time 5 min

Final Checks

- Perform QA/ RF Verify
- Perform QA/ Calculation Artefacts
- Perform QA/ Stability_Long Term Check (if available)
Time 30 min

2.5.5.4 CAN Bus Optical Controller

Material number: CAN module: 7388858

Required Time and Tools

Time

15 min

Tools

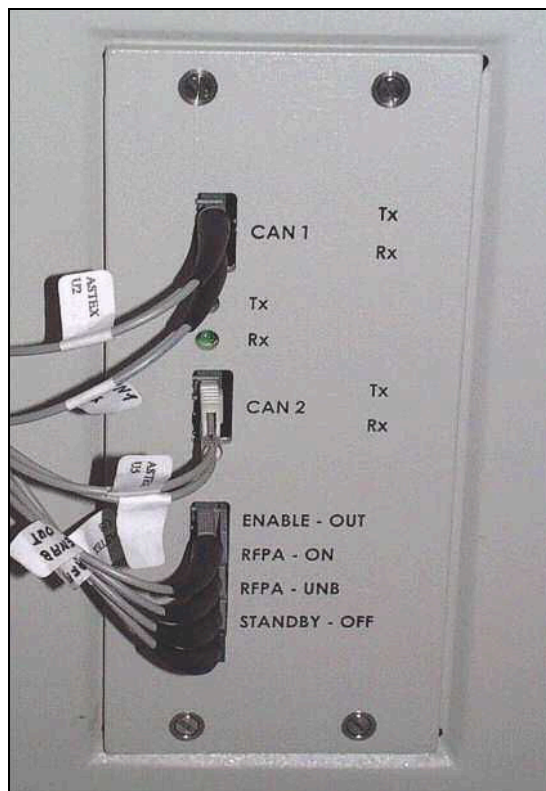
- 1/8 to 1/4 inch (3 mm to 6 mm) width blade-type screwdriver

Preliminary

1. Place the amplifier in OFF mode. Amplifier fans will remain "ON" for approximately one minute to allow for the amplifier to cool down.
2. After the fans stop, switch-off the mains circuit breakers (CB2, CB1 at amplifier PS deck and main circuit breaker in the mains box on top of CCA - in that order).

2.5.5.5 CAN Bus Controller Removal

Fig. 102: CAN Bus Controller Location



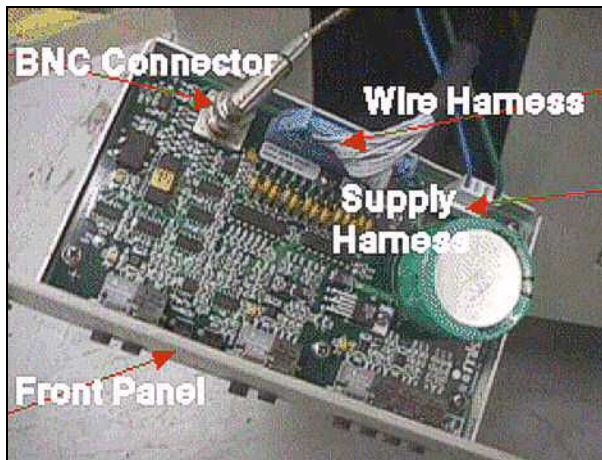
The CAN Bus Controller front panel is shown in (→ Fig. 102 Page 77) without the plastic protection cover.

To remove the CAN Bus controller, proceed as follows:

The CAN Bus controller is attached to the front panel of the power supply deck. It is secured in place with 4 captive slot-head screws that are permanently attached to the controller panel. Additionally, a plastic protection cover is secured via these screws.

- Turn the slot-head screws counter-clockwise (to the left) several turns until the plastic cover can be removed and the screws are finally free of the cabinet.

Fig. 103: CAN Bus Controller and Connectors



Once the controller panel screws are free, the assembly will drop forward and out of the power supply deck. The controller is connected to the amplifier via wire harness, separate RF BNC-type, and power cables. The harness and cable connections can be unplugged from the controller. The individual connectors are shown in (→ Fig. 103 Page 78).

2.5.5.6 CAN Bus Controller Installation

To install the CAN Bus controller, proceed as follows:

- The controller must be connected to the amplifier wire harness and separate the RF and power cables. Pass the harness and cables through the CAN Bus controller panel opening.
- Plug the harness and cables into the controller.
- Orient the controller panel so that the LED indicators are toward the left side of the power supply deck. Then attach the panel to the cabinet with the captive screws located at each corner of the panel. Don't forget to reattach the plastic protection cover.

2.5.5.7 Final Checks

- Switch ON the main circuit breaker on top of CCA, CB1 and CB2 at the amplifier PS deck - in that order.

2.5.6 Temperature Probe Replacement

Material number: Temperature probe 10018752

2.5.6.1 Required Time and Tools

Time

45 min

Tools

- 1/8 to 1/4 inch (3 mm to 6 mm) width blade-type screwdriver
- #2 cross-type screwdriver
- 3/4 or 11/16 Inch wrench (19 mm or 18 mm) (depending on which nut was used)

2.5.6.2 Preliminary

1. Place the amplifier in OFF mode. Amplifier fans will remain "ON" for approximately 1 minute to allow for the amplifier to cool down.
2. After the fans stop, switch-off the line circuit breakers (CB2, CB1 at the amplifier PS deck and main circuit breaker in the mains box on top of CCA - in that order). Wait 6 minutes to allow for the amplifier high voltage discharge.



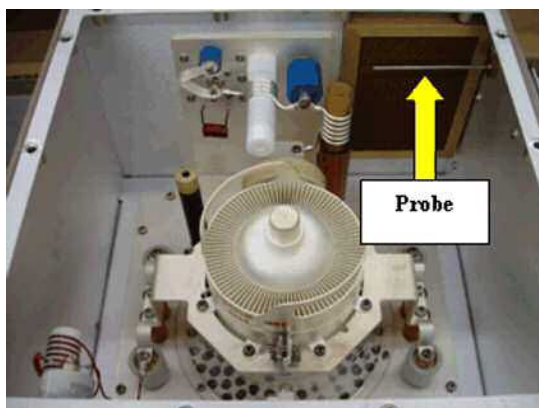
WARNING

High voltage may remain at the tubes and other components in the compartment. If ignored, death or serious physical injury can occur.

- » Do not touch anything in the compartment until the high voltage is discharged using the discharge rod (→ Fig. 78 Page 61) in the following step!

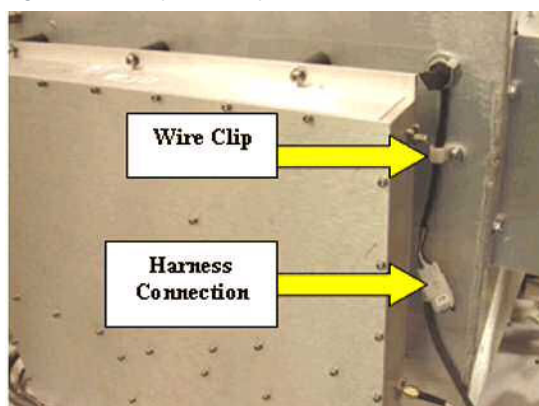
2.5.7 Replacement Procedure - Original Location

Fig. 104: Temperature probe - original location



- Using a blade-type screwdriver to remove the top and right side covers of the RF deck.
- Using the #2 cross type screwdriver, remove the tube compartment cover.
- The temperature probe will be visible in back inside the tube compartment (→ Fig. 104 Page 79).

Fig. 105: Temperature probe - old connection



- Remove the wire clip located on the side of the RF deck and disconnect the harness (→ Fig. 105 Page 79).

- Remove the nut that secures the temperature probe to the compartment wall. The part can now be removed from the RF deck.
- Reverse the previous steps to install the new temperature probe

2.5.8 Replacement Procedure - New Location

Fig. 106: RF output



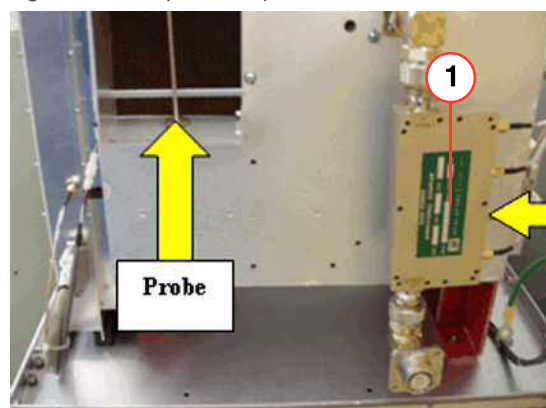
- The RF output connection is located on the rear panel of the amplifier. This connection is secured to the panel with a square bracket and 8 screws (→ Fig. 106 Page 80).

- Remove the 8 screws and bracket from the rear panel.
- Using a blade-type screwdriver remove the top, left and right side covers of the RF deck.
- Disconnect the green ground wire that is secured to the inside of the rear panel.
- Remove the 7 remaining screws from the lower edge of the rear panel. Start with the ones on each side, then remove the 5 from the bottom.



These 5 screws are the last screws holding the panel to the amplifier. Make sure to keep the panel from falling over or dropping while the screws are being removed. Damage to the panel could occur.

Fig. 107: Temperature probe - new location



(1) RF output assembly

- After removal of the rear panel, the temperature probe will be visible (→ Fig. 107 Page 80).

- Use caution when working around the RF output assembly- The assembly is suspended from the RF deck and the connections are easily damaged.

Fig. 108: Temperature probe - new connection



- (1) Nut
- (2) Wire clip
- (3) Harness connection

- Remove the wire clip located on the side of the RF deck and disconnect the harness (→ Fig. 108 Page 81).
- Remove the nut that secures the temperature probe to the RF deck air exhaust port. The part can now be removed from the RF deck (→ Fig. 108 Page 81).

- Install the new temperature probe.
- Secure the rear panel to the RF deck using 7 screws. (Do not secure these screws completely).
Use caution when lining up the RF output connection into the hole of the rear panel so not damage will occur.
- Reconnect the ground wire to the rear panel.
- Reinstall the side RF deck panels while leaving the screws loose.
- Reinstall the top cover. All screws should be started loose before they are tightened down to make it easier to get all screws installed. Use caution so that none of the screws is cross-threaded.
- Once the top cover screws are started, all cover screws can be secured.
- Reinstall the RF output bracket by starting with the 4 inner screws and finishing with the 4 outer screws.

2.5.8.1 Final Checks

- Switch ON the main circuit breaker in the mains box on top of CCA, CB1 and CB2 at the amplifier PS deck - in that order.
- Perform an MR image measurement.

2.5.9 Air Filter Replacement

Material number: 7389807

Air filters located on the RF and power supply panels are attached to the panels with easily removable machine screws. Use a cross-blade or flat blade screwdriver, as appropriate, to remove and install the filters.

The filter on the RF deck is friction-fit to the panel with “J” hooks. To remove it, place your hands at each side of the filter and firmly push it up. It will travel approximately 1/2 inch (1.25 cm). Then pull it straight out to disengage the “hooks”.

Version 04

- Chapter "RFPA" section "RFPA 35 kW replacement (RF Deck and/or PS Deck) Deck recombination (→ Deck Recombination / Page 75) check of fan power plug connection added; graphic Ground wire detail updated, CHARM MR_00451966 and general editorial changes.

Version 05

- All chapters - minor editorial changes.

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